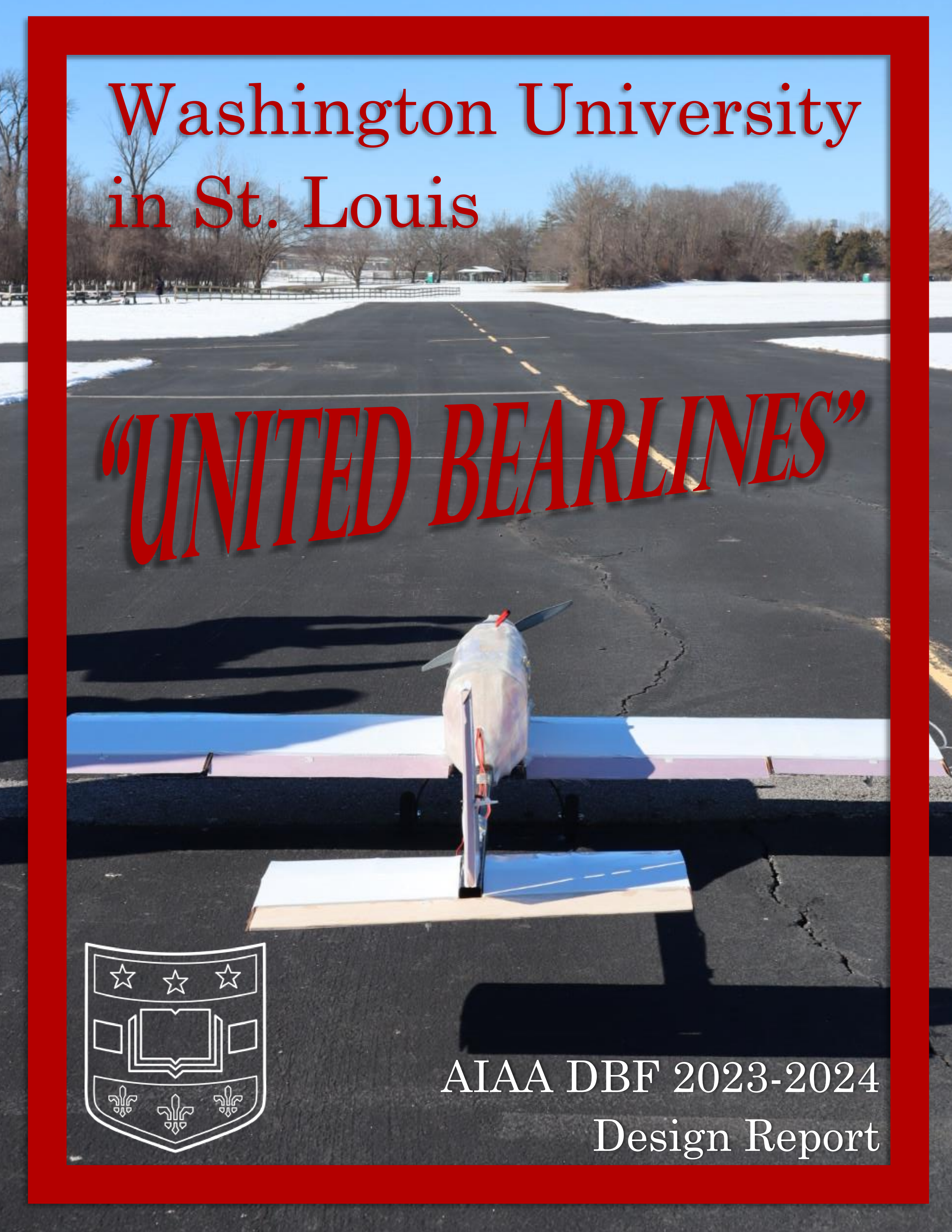
Washington University in St. Louis

"UNITED BEARLINES"



AIAA DBF 2023-2024

Design Report

Contents

Nomenclature

Abbreviations

AIAA	American Institute of Aeronautics and Astronautics
AoA	Angle of attack
AR	Aspect ratio
AUW	All-up weight
CA	Cyanoacrylate
CAD	Computer-aided design
CBDM	Component drag build-up method
CF	Carbon fiber
CFD	Computational Fluid Dynamics
CG	Center of gravity
DBF	Design/Build/Fly
ESC	Electronic Speed Controller
FEA	Finite Element Analysis
GM	Ground Mission
HQRS	Cooper-Harper handling qualities rating scale
HT	Horizontal tail
Km_sim	Kinematic mission simulator
LiPo	Lithium Polymer
M1	Mission 1
M2	Mission 2
M3	Mission 3
NACA	National Advisory Committee for Aeronautics
NiCad	Nickel Cadmium
NiMH	Nickel-metal hydride
NP	Neutral Point
RC	Radio-controlled
RPM	Revolutions per minute
SM	Static margin
TOFL	Takeoff field length
T-O	Takeoff
VLM	Vortex lattice method
VT	Vertical tail
WU	Washington University
WUDBF	Washington University in St. Louis Design/Build/Fly
XPS	Extruded Polystyrene

Symbols

α	Angle of attack (deg)
λ	Taper ratio
δ	Aileron deflection (deg)
ν	Kinematic viscosity (m^2/s)
ρ	Density (kg/m^3)
A	Characteristic reference area (m^2)
b	Span (m)
c	Chord (m)
C_D	Drag coefficient
C_L	Lift Coefficient
F	Drag force (kgf)
I_T	Optimal tail arm length
MAC	Mean aerodynamic chord (m)
Re	Reynolds number
S	Wing area (m^2)
T	Thrust (kgf)
T/W	Thrust-to-weight ratio
u	Velocity (m/s)
$\mathbf{V}$	Volume (m^3)
V_∞	Freestream velocity (m/s)
v_{max}	Max cruise velocity (m/s)
W	Weight (kgf)
W/S	Wing loading (kg/m^2)
X	x-location (m)

1 Executive Summary

The Washington University in St. Louis Design/Build/Fly (WUDBF) 2023-24 Design Report describes the design, testing, and manufacturing of *United Bearlines*, the team's entry in the American Institute of Aeronautics and Astronautics (AIAA) 2024 Design-Build-Fly competition. For the competition, the aircraft must be able to complete three flight missions and one ground mission. The first flight mission (M1) is a staging flight that verifies the function of the aircraft and demonstrates that the aircraft can carry the Crew payload. The second mission (M2) is the medical transport flight. In this flight, the aircraft must carry the Crew, EMTs, a Patient on a gurney, and a Medical Supplies Cabinet while traversing the course as fast as possible. The third mission (M3) is an urban taxi flight in which the aircraft must carry the Crew and Passengers while completing as many laps as possible around the course in a five minute window. The ground mission (GM) illustrates the aircraft's ability to change mission configurations.

United Bearlines has been designed to meet all mission requirements while maintaining a high level of reliability and minimizing risk. The aircraft consists of a conventional, single motor, bow tricycle landing gear, low-wing design. These characteristics were chosen during the conceptual design phase in response to the stated requirements. The bow tricycle configuration was chosen for its structural stability and to maximize the short 20 ft takeoff field length (TOFL) requirement. The low-wing configuration was chosen because it allows for quick passenger installation and does not interfere with the required loading hatches, which are critical for all missions. A sensitivity analysis performed during the conceptual design phase indicated that a M2 payload of 1.965 kg and an M3 payload of 24 passengers with a 100 Wh battery was optimal given the size and scoring constraints and WUDBF's desire for a low risk design.

During the preliminary design phase, the aircraft was further refined via detailed aerodynamic and propulsion design. Lift, drag, and stability analyses were performed to size the aircraft and set the outer geometry. The wing span was set at 1.34 m with a maximum wing loading of roughly 12 kg/m^2. The payload placement in M2 was chosen such that the change in overall center of gravity (CG) between missions is minimized. A propulsion system was then selected that gives the aircraft a 2.18 thrust-to-weight ratio and a predicted maximum cruise velocity of 34.08 m/s.

During the detail design phase, the internal structure of the aircraft was defined. The interior of the fuselage was designed primarily for optimizing access to the M2, M3, and GM payloads. The passenger, pilot, and medical inserts were designed to fit securely inside the fuselage and are held in place with Velcro. The medical payloads are secured to the inserts with a locking plate and cotter pins. Two hatches were designed to allow for easy loading and unloading during all missions.

Based on subsequent performance simulations, the aircraft was predicted to be able to take off in under 16 ft in its heaviest configuration, well below the 20 ft requirement. The aircraft was predicted to be able to complete M2 in 1.97 min, well within the time limit and yielding a score of 1.611. For M3, the aircraft was predicted to fly for a total of 4.47 min while completing 7 scoring laps and carrying 24 passengers, yielding a predicted score of 2.196. These predicted flight times aligned well with the endurance estimates made during the conceptual design phase. Adding on the predicted M1 and GM scores, *United Bearlines* is estimated to score competitively with a total score of 5.4 out of a possible 7. Thorough ground and flight testing confirmed that the aircraft could meet all predicted performance goals.

2 Management Summary

2.1 Organization

To facilitate efficient communication and work, WUDBF is organized into an Administrative Branch and a Technical Branch, both of which are overseen by a Faculty Advisor. Information and decisions flow downward from the Presidents to the Team Leads and then ultimately to the General Members. The Administrative Branch is responsible for resource allocation, public relations, and recruiting. The Administrative Branch consists of the Administrative President, the Finance Lead, and the Communications Lead. The Technical Branch is responsible for the design and manufacture of the aircraft. The leadership structure of WUDBF is described in Fig. 2.1. The Technical Branch consists of the Technical President and six Teams as described in Table 2.1. The number of General Members assigned to each Team is based primarily upon student interest, but it also depends on Team needs.

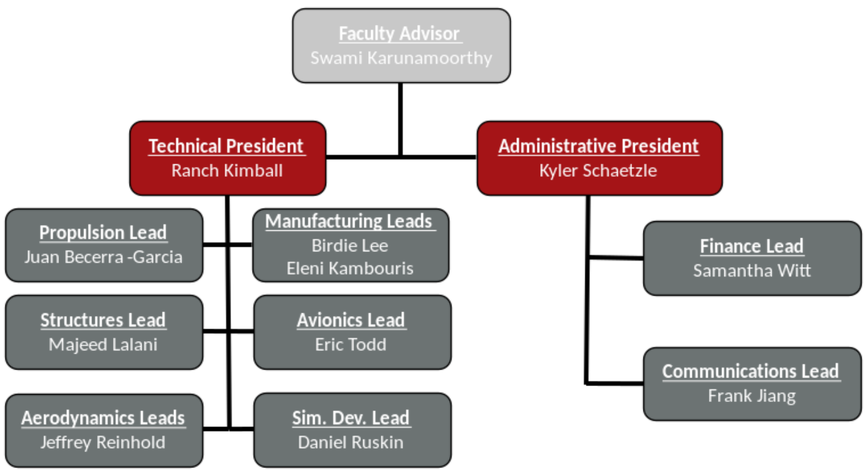


Figure 2.1: WUDBF Team hierarchy.

Table 2.1: Team roles and skills.

	Aerodynamics	Structures	Manufacturing	Avionics	Propulsion	Simulation Development
Roles	• Design of the exterior geometry of the aircraft • Aircraft sizing	• Design and modeling of the aircraft structure	• Assembly of the aircraft • Materials testing	• Installation of electrical components • Flight controller programming	• Design of the propulsion package • Motor testing	• Development of simulation tools • Sensitivity analysis
Skills	• Aircraft stability and control optimization • Computational Fluid Dynamics (CFD) analysis	• Computer-aided Design (CAD) • Structural Finite Element Analysis (FEA) • Mass properties analysis	• Laser cutting • 3D printing • Machining	• Electrical Design • Soldering and wiring	• Propulsion theory and analysis • eCalc • Remote Control (RC) Benchmark	• MATLAB • Simulink • Python

2.2 Schedule

WUDBF uses a Gantt chart to ensure the team stays on track to meet major deadlines. The Gantt chart for the 2023-2024 competition is shown in Fig. 2.2 below.

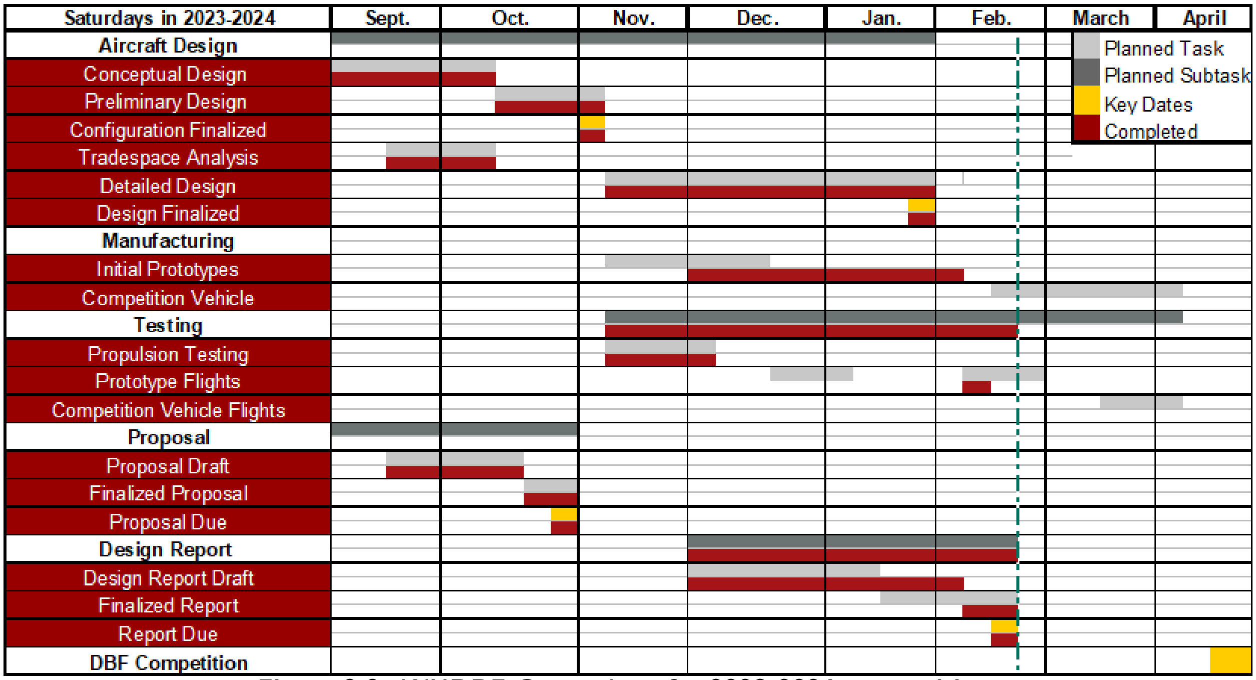


Figure 2.2: WUDBF Gantt chart for 2023-2024 competition.

3 Conceptual Design

For the 2023-24 Design/Build/Fly (DBF) competition, the AIAA has set the objective to design and build an urban air mobility aircraft. The aircraft must be capable of transporting a medical supply cabinet and medical personnel, and civilian passengers while maintaining a compact design. Upon receipt of the competition rules, WUDBF began the conceptual design phase starting with identification of key mission requirements and translation of those mission requirements into design requirements, followed by a scoring sensitivity analysis and configuration selection via a series of decision matrices.

3.1 Mission Requirements

The 2024 competition consists of four missions; one ground mission and three flight missions. The same course is used for all flight missions and is shown in the figure below. Both the flight missions and the ground mission follow the same setup procedure. The aircraft must enter the staging area in its parking configuration, then be deployed into its flight configuration and loaded with the mission payloads within five minutes.

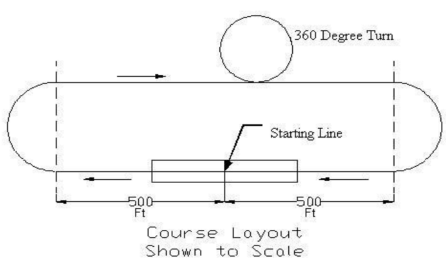


Figure 3.1: Competition course layout

3.1.1 Mission 1 - Staging Flight

The payload for Mission 1 is the Crew only. The aircraft must takeoff within 20 feet, and complete 3 laps of the course within a five-minute window. Landing may occur outside the window, but the aircraft must successfully land to get a score. Mission 1 scoring is calculated as follows:

$$\mathrm{M1} = 1.0 \tag{3.1}$$

The maximum score for mission one is **1.0 point** awarded for completion.

3.1.2 Mission 2 - Medical Transport Flight

The payload for the Medical Transport Flight consists of the Crew, the EMT, the Patient on gurney, and the Medical Supply Cabinet. The aircraft must take off within 20 feet, and complete three timed laps of the course within a five-minute window. A lap is counted when the aircraft crosses the start/finish line. Landing may occur outside the window, but the aircraft must successfully land to get a score. Mission 2 scoring is calculated as follows:

$$\mathrm{M2} = 1.0 + \frac{(payloadweight/time)_{\mathrm{WUDBF}}}{(payload\_weight/time)_{\max}} \tag{3.2}$$

Where *payload weight* is the weight of the medicine supply cabinet, and *time* is the time it takes to complete three laps. The WUDBF and max subscripts denote the raw scores achieved by WUDBF and the team that scored highest on M2, respectively. A total of **2.0 points** are possible on M2, with one coming from completion and a second from performance.

3.1.3 Mission 3 - Urban Taxi Flight

The payload for the Urban Taxi Flight is the Crew and Passengers. The aircraft must take off within 20 feet, and complete as many laps of the course as possible within a five-minute window. A lap is counted when the aircraft crosses the start/finish line. Landing may occur outside the window, but the aircraft must successfully land to get a score. Mission 3 scoring is calculated as follows:

$$\mathrm{M3} = 2.0 + \frac{(\#laps * \#passengers/batterycapacity)_{\mathrm{WUDBF}}}{(\#laps * \#passengers/batterycapacity)_{\max}} \tag{3.3}$$

Where *#laps* is the number of laps completed, *#passengers* is the number of passengers carried, and *battery capacity* is the propulsion battery capacity in Wh. The WUDBF and max subscripts denote the raw scores achieved by WUDBF and the team that scored highest on M3, respectively. The maximum score on Mission 3 is **3.0 points**, with two points awarded for completion and a third for performance.

3.1.4 Ground Mission - Configuration Demonstration

The Ground Mission serves as a timed ground demonstration of the aircraft's Mission 2 and 3 capabilities. The Ground Mission begins with the aircraft in the parking configuration with no payloads installed. The Crew, Medical Supply Cabinet, EMT, Patient and gurney, and Passengers, as well as any floor inserts, are placed in the triage area next to the parking spot. The mission is divided into three phases. In the first, the mission official will say "GO" and begin the timer. The assembly crew member will place the aircraft in flight configuration and load the M2 payloads, securing all hatches. The official will then pause the timer. Hatches and flight controls will then be verified. In the second phase, the official will again say "GO" and the assembly crew member will remove the M2 payloads and insert the passengers, again securing all hatches. The timer will pause, and hatches and flight controls will then be verified. In the third phase, the official will say "GO", and the assembly crew member will unload the passengers and return the aircraft to its parking configuration, stopping the time for the mission. Scoring is as follows:

$$\text{GM} = \frac{(time)_{\text{min}}}{(time)_{\text{WUDBF}}} \tag{3.4}$$

Where *time* is the total Ground Mission Time. The WUDBF and min subscripts denote the raw scores achieved by WUDBF and the team with the lowest GM time, respectively. The maximum achievable score for the Ground Mission is **1.0 points** based on performance.

3.1.5 Total Competition Score

Mission 1, 2, 3, and the Ground Mission scores are combined to give the total mission score:

$$\text{Total Mission Score} = \text{M1} + \text{M2} + \text{M3} + \text{GM} \tag{3.5}$$

The maximum achievable Total Mission Score is **7.0 points**, with four awarded for completion and three for performance. The final Competition Score is calculated as follows:

$$\text{Competition Score} = \text{Total Mission Score} * \text{Report Score} + P \tag{3.6}$$

Where P is the participation score, with one point awarded for attending the fly-off, a second for completing tech inspection, and a third for attempting a flight mission.

3.1.6 General Mission Requirements

In addition to the requirements specific to each mission, there are also a number of other rules of note that apply to the competition aircraft during all missions:

- The aircraft's wingspan cannot exceed five feet in the flight configuration.
- The aircraft must fit inside a parking spot 2.5 feet wide in its parking configuration.
- If the wingspan exceeds 2.5 feet, the aircraft must be able to fit without removing any components except fasteners, which must remain in the same location in both configurations.
- The gurney must be at least the same width and length as the Patient with a minimum height of 1.5 inches.
- The Medical Supply Cabinet shall have a minimum width and length of 3.00 inches and a minimum height of 3.50 inches.
- Inserts may be used to secure payloads. The passengers may not touch each other or any part of the aircraft except the inserts.
- There must be a solid bulkhead between the crew and payload compartments, and the crew must have an unobstructed view out the front of the aircraft.
- The aircraft may utilize either Nickel Cadmium (NiCad)/Nickel Metal Hydride (NiMH) OR Lithium Polymer (LiPo) batteries for propulsion, but the total propulsion power total stored energy cannot exceed 100 watt-hours.
- The flight missions must be flown in order; a new mission cannot be flown until the team has obtained a successful score for the preceding mission.

3.2 Translation to Design Requirements

Based on the mission requirements outlined above, WUDBF formulated design requirements for each mission. These design requirements are broken down in Table 3.1.

Table 3.1: Translation of mission requirements into design requirements.

Mission	Mission Requirement	Design Requirement
All	20 ft TOFL	Low wing loading (W/S), high maximum lift coefficient ($C_{L,max}$), high thrust-to-weight ratio (T/W)
M2	Minimize lap time	High cruise velocity, high T/W, low drag
	Maximize Medical Supply Cabinet weight	Large wing area, sufficient internal volume
M3	Maximize laps	High cruise velocity, high T/W, low drag, high propulsive efficiency, high battery capacity
	Maximize number of passengers	Large wing area, sufficient internal volume
	Minimize battery capacity	High propulsive efficiency, low battery capacity
GM	Minimize loading time	Optimized payload storage configuration, easy access to internal storage, easy switching between configurations, limited number of passengers
General	Complete every mission	Low risk design, high stability
	Max propulsion energy of 100 watt-hours	High efficiency, constrains maximum propulsion power
	Max wingspan of 5 ft	Constrains wingspan
	Aircraft in parking configuration must have width within 2.5 ft	Constrains wingspan and tail span, and/or encourages flexible wing and tail adjustment

3.2.1 Analysis of Design Requirements

TOFL: Significant design effort must be made to follow the short takeoff limit of 20 ft for every mission. The aircraft needs to have a high $C_{L,max}$, a low W/S, and a high T/W. These elements are critical for M2 and M3, where heavier payload is encouraged.

Mission 2: Higher scores in M2 are obtained by maximizing Medical Supply Cabinet weight while minimizing lap time. This leads to contradicting design considerations. To increase the supply cabinet weight, the aircraft must increase its wing area to support a realistic wing loading. Increasing the wing area leads to more drag, causing a decrease in cruise velocity and thus hurting lap time. Therefore, a balance must be struck between maximizing payload and maximizing cruise velocity.

Mission 3: Higher scores in M3 are obtained by maximizing laps, maximizing the number of passengers, and minimizing battery capacity. Maximizing laps requires better endurance, which is done by increasing battery capacity. Thus, the number of laps is directly proportional to battery capacity. However, as shown in Section 4.1.3, increasing battery capacity also reduces mission score. For simplicity, the positive and negative effects of increasing battery capacity on mission score were considered to be approximately equal. Keeping this in mind, then the most important aspects of a high M3 score are increasing cruise velocity to maximize laps and increasing payload through the number of passengers. This leads to similar concerns to the ones noted in the analysis of M2 above. Once again, the design must balance between maximizing payload and maximizing cruise velocity.

Ground Mission: Minimizing Ground Mission time requires the aircraft to have hatches and doors that facilitate easy loading/unloading of the payloads for M2 and M3. GM affects M3 because loading/unloading time increases when the number of passengers increases, so the number of passengers in M3 may want to be decreased to help minimize GM time. The aircraft must also be designed for swift switching between flight and parking configurations.

Minimal Risk: Though mission performance is an important part of the competition score, over half of the mission points are awarded for completion. As such, the most important aspect of the aircraft's design is its overall stability and reliability.

Constraints: The 100 Wh battery limit imposes a limit on the power of the propulsion system, meaning that high efficiency is a must. The wingspan is constrained by the 5 ft wingspan limit as well as by the parking configuration width limit of 2.5 ft. This parking configuration limit also constrains the tail span, and may require the wings and/or tail to be flexible enough for readjustment between configurations.

3.3 Scoring Model and Sensitivity Analysis

Considering the design requirements above, WUDBF determined that no obvious solution presented itself to the design problem. Maximizing cruise velocity to decrease lap times in M2 and increase the number of laps in M3 would lead to a smaller, faster aircraft. However, maximizing payload in M2 and M3 would generate a larger, slower aircraft. And while maximizing the number of passengers would help the M3 score, it would also hurt loading/unloading times in GM. Given these conflicting strategies, WUDBF designed a scoring model and conducted a sensitivity analysis to determine which factors would contribute most to the total score.

3.3.1 Scoring Model

To create the scoring model, WUDBF swept through different values of five independent variables: Medical Supply Cabinet weight, number of passengers, battery capacity in M3, maximum cruise velocity in M2 and M3, and wing area. These variables were chosen because they could be utilized, along with other assumed values, to estimate every mission score. Thus, WUDBF calculated five-dimensional matrices for each mission, containing a score for each possible combination of values. Sweeping through these five variables also allowed for calculations of all mission constraints and self-prescribed constraints, which included minimums/maximums for takeoff length, wing loading, power, and endurance. These constraints were applied to the mission score matrices, eliminating all matrix positions corresponding to combinations of values that did not satisfy every constraint. Justifications for all assumed values and self-prescribed constraints are provided in Table 3.2.

Table 3.2: Table of key assumptions and justifications for the scoring model.

Assumption	Justification
Drag Coefficient ($C_{D,min}$) is 0.08.	Based on previous WUDBF aircraft testing and preliminary CFD analysis.
C_L is 0.96.	From pilot interviews, a Lift/Drag ratio of 12 is typical for RC aircraft.
Wing area varies between 0.1 and 0.5 m^2.	An aircraft with a wing area of 0.1 m^2 is unlikely to take off with the required payload, and an aircraft with a wing area of 0.5 m^2 is unlikely to satisfy the 5 ft wingspan limit.
The maximum wing loading is 11 kg/m^2.	Test pilot feedback on previous aircraft suggests a wing loading heavier than 11 kg/m^2 will make the aircraft very difficult to control.
The maximum power is 2000 W.	A 2000 W motor will deplete a 100 Wh battery in 3 minutes, and therefore will likely not score well on the endurance mission, if it is able to complete 3 laps at all.
The maximum number of passengers is 50.	It is unlikely that the aircraft could fit more than 50 passengers.
The maximum cruise velocity is 45 m/s.	Test pilot feedback on previous aircraft suggests a cruise velocity greater than 45 m/s will make the aircraft very difficult to control.
Lap length is approximately 2500 feet.	Based on the course diagram.

After constraints were applied, the mission score matrices were normalized by the highest score in each mission to simulate the competition rules. The automatic points granted for completing each mission were not included to increase contrast between the mission scores. Then, the score matrices were summed to create a total score matrix. The five input values corresponding to the maximum total score were used to generate initial design targets, displayed in Table 3.3.

Table 3.3: Initial design targets.

Design Specifications	Initial Target Value
Wing Area	0.36 m^2
M2 Medical Supply Cabinet Weight	1.88 kg
M2 All-up weight (AUW)	3.60 kg
M2/M3 Max Cruise Velocity	27.5 m/s
Number of Passengers	24
M3 Battery Capacity	32.0 Wh
M3 Endurance	5 min
M3 Number of Laps	7
Power	575.8 W

3.3.2 Sensitivity Analysis

The design targets were used to guide the initial conceptual design and were further refined in the preliminary design phase. Using the optimal point of variables corresponding to the maximum total score, a sensitivity analysis was done to see how deviations in each parameter would alter the total score. Understanding how slight alterations in inputs affect the total score would help guide future design decisions. Results of the analysis are shown in Fig. 3.2.

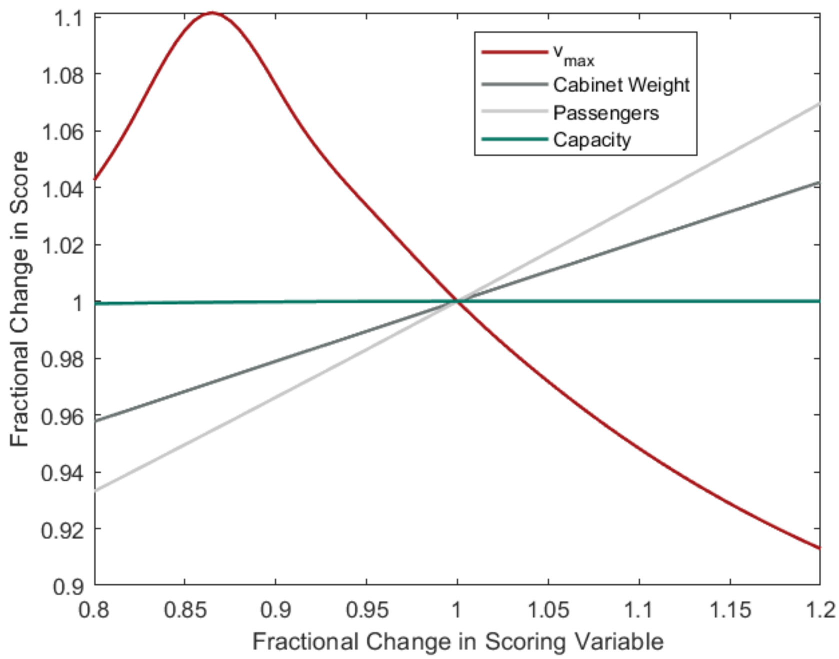


Figure 3.2: Sensitivity analysis results.

One observation from the sensitivity analysis is that changes in battery capacity do not significantly impact total score. As explained in Section 3.2.1, this is due to M3 score being both multiplied by battery capacity and divided by battery capacity, effectively canceling out the impact of battery capacity. The sensitivity analysis assumed the number of laps to be a continuous variable, even though in reality it is discrete. This was done since detailed aircraft parameters were not yet decided. Treating laps as a continuous variable allowed us to see how small changes in the assumptions affected aircraft performance. Given the minimal impact of battery capacity, WUDBF decided to use a battery capacity approaching the upper limit of 100 Wh for all missions, including M3. This fell in line with the battery capacity used in previous WUDBF aircraft. Another observation is that both M2 payload (Medical Supply Cabinet weight) and M3 payload (number of passengers) are positively correlated with total score, meaning that the final design should aim to maximize payloads. Since maximum payload is desired by the scoring model, then it makes sense that lower cruise velocities correlate with higher score in the sensitivity analysis. The model clearly encourages building a larger, slower aircraft instead of a smaller, faster aircraft. It should also be noted that the slope of the Passengers line is larger than that of the Cabinet Weight line. This indicates that small deviations in the number of passengers have a greater effect on total score than small deviations in the Medical Supply Cabinet weight.

3.4 Configuration Selection

Once initial design targets were determined, decision matrices were then used to select the overall aircraft configuration as well as the configuration of major subsystems. All decision matrix weights were determined based on design goals and competition requirements. Individual criterion scores were then determined based on team knowledge and experience from previous competitions. The individual criterion scores were multiplied by the weights and summed together to determine the total weighted score for each configuration. A selection was then made by comparing the total weighted score.

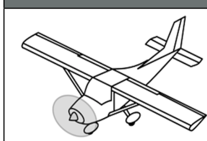
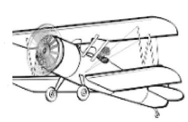
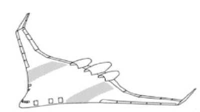
3.4.1 Configuration Selection

WUDBF first considered three overall aircraft configurations including the monoplane, biplane, and blended body configurations. Key design criteria included aerodynamic efficiency, stability and control, wing loading, and manufacturability. The highest weights were assigned to flight speed and stability and control as speed is important to earning points, and stability is a main driver of overall aircraft design. Using these criteria, the following configurations were compared in Table 3.4.

- **Monoplane**: Provides good aerodynamic efficiency, stability and control, and is likely the most feasible configuration. However, available wing area within the span limitation is the lowest of the considered configurations.
- **Biplane**: Potentially allows for more lift within the span limitation by maximizing available wing area. However, the increased drag and added structural weight associated with the configuration decreases efficiency.
- **Blended Body**: Allows for increased wing area while still maintaining relatively good aerodynamic efficiency. However, the complex geometry of the blended body reduces design and manufacturing feasibility. Additionally, the blended body configuration would have limitations in regard to stability and control.

Translating these judgments to the decision matrix resulted in Table 3.4 below:

Table 3.4: Selection matrix primary configuration.

		Score		
				
Selection Criterion	**Weight (%)**	Monoplane	Biplane	Blended Body
Flight Speed	25	3	1	5
Drag	15	4	1	5
Wing Loading	15	4	5	3
Stability and Control	25	5	2	5
Manufacturability	20	5	4	1
	Weighted Score	4.2	2.5	3.9
	Rank	1	3	2

Ultimately, the monoplane configuration was selected due to its superior efficiency over the biplane and easier manufacturability compared to the blended body.

3.4.2 Wing Placement

WUDBF next considered wing placement by comparing high-, mid-, and low-wing configurations. Takeoff capability and ease of loading were weighted highest followed by stability, flight speed, and maneuverability.

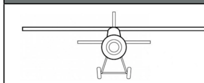
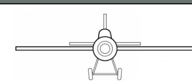
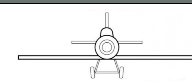
- **High-Wing**: Possesses the best stability and control due to its tendency to self-correct. Also is the most feasible configuration since WUDBF used it successfully in the past. However, access to the fuselage from above is restricted, increasing payload loading complexity and GM time.

- **Mid-Wing**: Has stability and control characteristics between that of the low- and high-wing configurations. However, this configuration necessitates that the structural support for the wing runs through the center of the fuselage. This results in a decreased payload capacity and potentially increased payload loading complexity because the bar might conflict with either the battery or the payload mass.

- **Low-Wing**: Because the low-wing configuration allows for easy access from above, it has superior payload loading capability. However, it is the least stable configuration due to the CG being above the wing. An unstable configuration would require constant maneuver corrections, ultimately slowing down the aircraft.

The resulting decision matrix is shown in Table 3.5 below:

Table 3.5: Selection matrix wing placement.

		Score		
				
Selection Criterion	Weight (%)	High Wing	Mid Wing	Low Wing
Maneuverability	10	3	4	5
Stability and Control	20	5	4	2
Takeoff Length	30	3	4	5
Ease of Landing	25	3	2	3
Flight Speed	15	3	4	5
	Weighted Score	3.4	3.5	3.75
	Rank	3	2	1

Ultimately the high wing was selected because it was the most stable orientation, and its manufacturing problems were deemed simple enough to overcome.

3.4.3 Empennage Configuration

WUDBF then selected the empennage configuration out of the conventional tail, the T-tail, the H-tail, and the V-tail. The main design criteria for empennage configuration were determined to be stability and control, flutter, drag, wake interference during takeoff, and feasibility. Using these criteria, the following configurations were compared in Table 3.6.

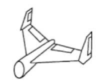
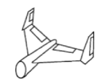
- **Conventional**: Simple to design and manufacture. Also possesses good stability and control and flutter characteristics.

- **T-tail**: Keeps the horizontal tail (HT) in undisturbed air, which would reduce the drag. However, the high-mounted HT can also generate low flutter speeds and asymmetric lift and yaw which induces high torsional loads. Additionally, having the HT at the tip of the vertical tail (VT) introduces additional complexity into the structure and manufacturing process.

- **H-tail**: Offers increased HT efficiency due to the endplates. However, the flutter speed of the H-tail is reduced due to the weight of the VTs being located at the tips of the HT. Additional control and structural complexity is introduced by having two VTs, decreasing feasibility.

- **V-tail**: Possesses good flutter characteristics and reduces the wake interference at takeoff. However, the V-tail control system is significantly harder to design and implement.

Table 3.6: Selection matrix empennage configuration.

		Score			
					
Selection Criterion	**Weight (%)**	Conventional	T-Tail	U-Tail	V-Tail
Drag	30	4	4	2	5
Stability and Control	40	3	4	5	4
Feasibility	30	5	3	2	2
	Weighted Score	3.9	3.7	3.4	3.7
	Rank	1	3	4	2

The conventional tail configuration was chosen to maximize manufacturability. Additionally, the conventional tail configuration is the design WUDBF has successfully employed in the past.

3.4.4 Landing Gear Configuration

The landing gear was then chosen based on ground maneuverability, drag, structural strength, feasibility, and takeoff capability. Using these criteria, the following configurations were compared in Table 3.7.

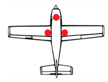
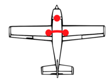
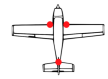
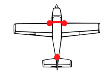
- **Tricycle**: Three struts or fairings independently connected to the fuselage. Provides good ground maneuverability and provides the option to allow the landing gear to retract into the fuselage. However, the structural strength of the front landing gear and takeoff capability of this configuration is a concern.

- **Bow Tricycle**: Three struts or fairings independently connected to the fuselage. Provides good ground maneuverability and provides the option to allow the landing gear to retract into the fuselage. However, the structural strength of the front landing gear and takeoff capability of this configuration is a concern.

- **Strut Tail-dragger**: Three struts or fairings independently connected to the fuselage. Provides good ground maneuverability and provides the option to allow the landing gear to retract into the fuselage. However, the structural strength of the front landing gear and takeoff capability of this configuration is a concern.

- **Bow Tail-dragger**: Two front wheels connected to the fuselage via a common strut or fairing and one rear wheel connected to the tail. Offers the same improved takeoff capability as the strut tail-dragger, while also providing better structural support. Ground maneuverability of this configuration is limited.

Table 3.7: Selection matrix landing gear configuration.

		Score			
					
Selection Criterion	**Weight (%)**	Tricycle	Bow Tricycle	Strut Tail-Dragger	Bow Tail-Dragger
Maneuverability	30	5	5	2	2
Stability and Control	25	4	4	4	3
Takeoff Length	25	2	3	3	5
Ease of Landing	10	4	5	3	2
Flight Speed	10	5	5	3	3
	Weighted Score	3.9	4.25	2.95	3.1
	Rank	2	1	4	3

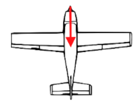
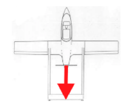
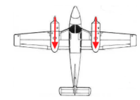
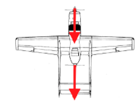
The Bow Tricycle configuration was chosen because it scored the highest by performing very well in ground maneuverability and feasibility compared to the other designs.

3.4.5 Propulsion Configuration

The propulsion system was chosen based on its efficiency, thrust, weight, component interference, and drag. The following propulsion configurations were considered and the constructed decision matrix is shown below in Table 3.8.

- **Tractor**: Single motor on the front of the aircraft. Provides the highest efficiency, lowest weight and drag, and is the most feasible. Ground clearance could be a concern depending on the required propeller size and height of front landing gear.
- **Pusher**: Single motor aft of the fuselage. Provides low weight and drag but has a decreased ground clearance (especially during rotation) and low efficiency due to fuselage interference.
- **Twin**: Two motors which are mounted on the wings. Provides improved ground clearance due to reduced propeller size and reasonable efficiency. Also offers torque balancing and is reasonably feasible. However, drag is increased resulting from the placement of the motors on the leading edge of the wing. Weight is also a concern because of the additional electronics and structural strength needed to support the second motor.
- **Push/Pull**: One motor at the front and aft of the fuselage. Offers low weight and drag. However, is the least efficient due to propwash interference, has the same ground clearance concerns as the tractor and pusher configurations, and is the least feasible. Weight is also a concern because of the additional electronics and structural strength needed to support the second motor.

Table 3.8: Selection matrix propulsion configuration.

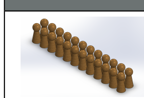
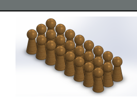
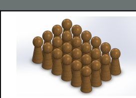
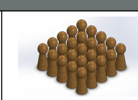
		Score			
					
Selection Criterion	**Weight (%)**	Tractor	Pusher	Twin	Push/Pull
Maneuverability	35	5	4	3	3
Stability and Control	25	3	3	5	4
Takeoff Length	15	5	5	3	3
Ease of Landing	10	3	3	5	1
Flight Speed	15	3	4	3	5
	Weighted Score	4	3.8	3.7	3.35
	Rank	1	2	3	4

The tractor configuration was ultimately chosen because it was thought to maximize efficiency while minimizing weight and drag. Additionally, the tractor configuration is the most feasible design and is one that WUDBF has successfully employed in the past.

3.4.6 Load Configuration

The load configuration was chosen based on its longitudinal weight distribution, lateral weight distribution, vertical weight distribution, ease of loading, and fuselage space. The following loading configurations were considered and the constructed decision matrix is shown below in Table 3.9.

Table 3.9: Selection matrix loading configuration.

		Score			
					
Selection Criterion	**Weight (%)**	2x12	3x8	4x6	6x4
Longitudinal weight distribution	20	2	4	3	5
Lateral weight distribution	20	3	4	3	4
Ease of Loading	35	4	2	4	1
Fuselage space	25	4	3	4	2
	Weighted Score	2.6	2.6	2.8	2.4
	Rank	2	2	1	4

The 4x6 configuration was ultimately chosen because it was thought to be easy to load, and is longitudinally stable compared to the other configurations.

3.5 Final Conceptual Design

The final conceptual design consists of a low-wing, monoplane, bow-tricycle aircraft with a conventional tail and a propulsion system in the tractor configuration. The aircraft is designed to carry an M2 payload of 1.880 kg and an M3 payload of 1.400 kg. The targeted max cruise velocity in M2 is 27.5 m/s. The mission 3 passengers will be configured in a 4x6 pattern.

4 Preliminary Design

During the preliminary design phase, the initial aircraft concept and configuration from the conceptual design phase were further developed. Key aspects of the design including the overall sizing, wing geometry, airfoil, and propulsion package were refined following several trade studies.

4.1 Design Methodology

The full design methodology employed by WUDBF is outlined in Fig. 4.1 below.

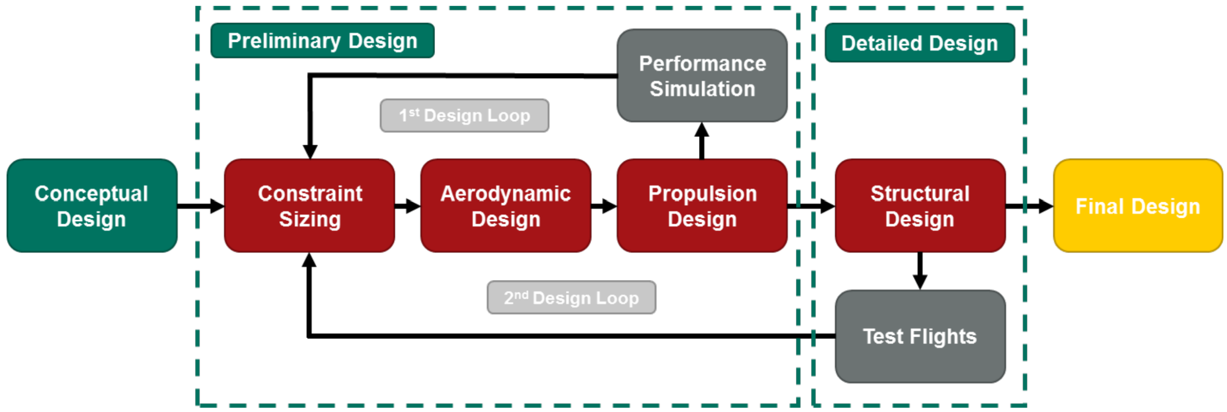


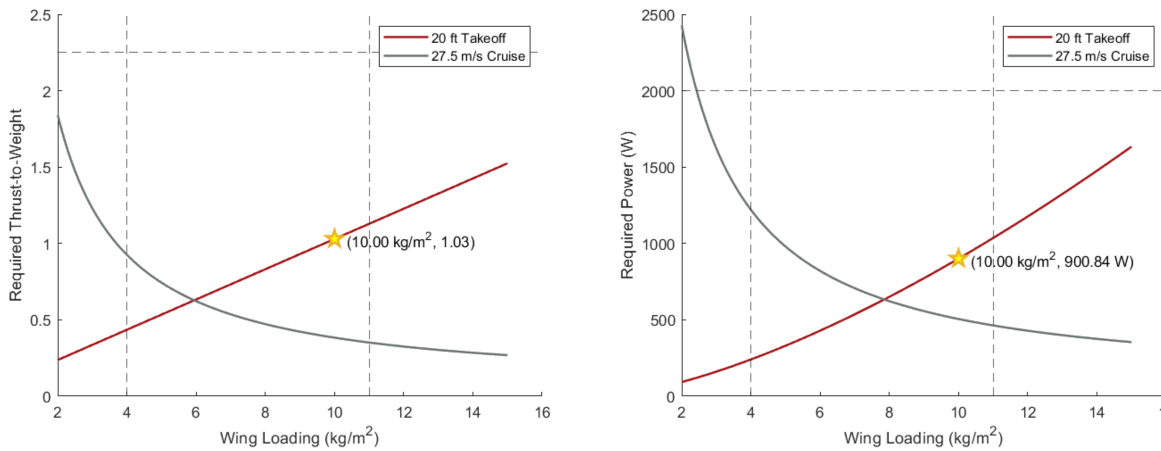
Figure 4.1: WUDBF design methodology.

For the preliminary design phase, WUDBF utilized an iterative process focused on the performance of the fully loaded aircraft (i.e. with the Supply Cabinet). This was done because 1) the AUW for M2 is the greatest out of the flight missions, and therefore the constraints will be more limiting, and 2) aircraft are typically designed for the fully loaded case. Initial design targets from the conceptual design phase were used as a starting point. A trade study was performed on the wing geometry to determine the overall size of the aircraft. The chosen wing design was then used to guide the aerodynamic design, which dictated the airfoil and tail sizing. The performance and stability characteristics of the aircraft were analyzed using the vortex lattice method in the analysis tool XFLR5. The drag polar of the aircraft was determined using induced drag data from XFLR5 and the component drag build-up method. Following the aerodynamic design, propulsion design took place using the target cruise, payload, drag polar, and endurance as starting points. The online tool eCalc was used to sort through a database of components to identify candidate propulsion system combinations. Data from eCalc was then used to select the optimal system, and performance was confirmed using WUDBF's time domain propulsion simulator written in MATLAB.

WUDBF employed two design loops to iterate the design. Following the first iteration, the preliminary design was tested using a kinematic mission simulator (Km_sim) developed by WUDBF. Performance estimates from Km_sim were then used to iterate the preliminary design. Once the design met desired performance goals in Km_sim, the design moved into the second design loop. Structural design took place to create a detailed CAD model of the aircraft. This model was used to build a prototype aircraft. Flight test data from the prototype were then used to inform changes to the preliminary design. Ultimately, the final design developed as the culmination of several iterations, with each iteration improving on performance and mission scores.

4.2 Constraint Sizing

An initial constraint sizing was performed by plotting the required thrust-to-weight ratio to achieve a 20 ft takeoff and 27.5 m/s cruise speed for a range of wing loadings using the equations developed in Gudmundsson [1]. These plots are shown in Fig. 4.2. The cruise speed of 27.5 m/s was one of the initial design targets determined by the scoring analysis in Section 3.3. Since the aircraft must be propeller driven using an electric motor, it is useful to convert T/W to required electrical power using initial estimates of weight, mechanical efficiency, and propulsive efficiency. This meant that the upper power constraint of 2000 W listed in Table 3.2 could be applied in the constraint sizing, along with an upper T/W constraint of 2.25.



(a) Required T/W vs. wing loading. **(b) Required power vs. wing loading.**

Figure 4.2: Constraint sizing plots.

It is clear from the rightmost plot that the upper power constraint is obeyed by almost all wing loading values. This allowed some flexibility in choosing a wing loading, as long as it was below our upper constraint of 11 kg/m^2.

Ultimately, a wing loading of 10 kg/m^2 was chosen for the design point as a conservative choice. In previous competitions, the final WUDBF aircraft has consistently exceeded the desired weight. Thus, designing an aircraft with a slightly higher wing loading allows for a larger structural weight margin (given that propulsion and payload weight are essentially constant).

From the chosen wing-loading of 10 kg/m^2, the minimum required power was set at about 901 W and the minimum required T/W ratio at about 1.03 (as indicated by the stars in the figures above).

4.3 Initial Weight Estimation

The AUW of the aircraft was estimated using a breakdown of individual components. The weight of the wing, HT, VT, and fuselage were estimated using previous aircraft, which were built with similar methods and materials. Battery, motor, servo, electronics, central carbon fiber (CF) rod, and landing gear weights were determined by weighing representative components already possessed by WUDBF. The M2 mass quantity was determined using the optimization program described in Section 3.3.1.

Table 4.1: Component weight breakdown.

Component	Weight [kg]
Propulsion Battery	0.438
Motor	0.400
Fuselage	0.285
Electronics	0.150
Avionics Battery	0.108
Wing Left	0.090
Wing Right	0.090
Rear Landing	0.090
Electronic Steering Control	0.080
Front Landing	0.040
Counterweight(M1)	0.250
M2 Payload	1.880
M3 Payload	1.400
Total M1	**2.021**
Total M2	**3.651**
Total M3	**3.063**

Based on this analysis, the predicted AUW of the fully loaded aircraft during M1 is 2.021 kg, during M2 it is 3.651 kg, and during M3 it is 3.063 kg.

4.4 Aerodynamic Performance

The aerodynamic design of the aircraft was iteratively improved to ensure that it is stable, controllable, and achieves maximum aerodynamic performance. XFLR5, MATLAB, and Ansys were the primary tools used for aircraft aerodynamic analysis. The size and shape of the main wing and stabilizers were driven by iterative static and dynamic stability analyses. Control surfaces and high lift devices were dimensioned to improve handling characteristics and minimize takeoff distance, respectively.

4.4.1 Main Airfoil Selection

In order to select the main airfoil, the predicted Reynolds number (Re) of the aircraft was first calculated using Equation 4.1 [1].

$$Re = \frac{c \cdot V_\infty}{\nu} \tag{4.1}$$

Where c is the mean aerodynamic chord [m], V∞ is the freestream velocity [m/s], and ν is the kinematic viscosity of air [m^2/s]. Kinematic viscosity was calculated from the average temperature in April (23 °C) and altitude (277 m) in Kansas City which yielded a kinematic viscosity of $1.498 \cdot 10^{-5}$ m^2/s. These conditions yielded a Reynold's number of 510,000.

Airfoil selection was then performed by selecting several different airfoils with maximum C_L/C_D near a Reynold's number of 500,000. Additionally, the search was restricted to airfoils with a minimum thickness of 10% and a maximum camber of 8% to ensure that the airfoil could accommodate structural and system integration and be reasonably manufactured with the processes available. Five promising airfoils were selected for comparison as shown in Table 4.2. Once these potential airfoils were identified, analysis of each airfoil was performed in

Table 4.2: Airfoil selection options [2].

Airfoil	Max CL/CD	Max. Thickness [% of chord]	Max. Camber [% of chord]
Clark X	102.7	11.7	3.3
Clark Y	98.7	11.7	3.4
NACA 1412	73.5	12.0	1.0
NACA 2412	87.3	12.0	2.0
USA-35B	106.0	11.6	4.1

XFoil and XFLR5 to determine which provided a favorable lift coefficient, lift-to-drag ratio, and drag bucket at a Reynold's number of 510,000.

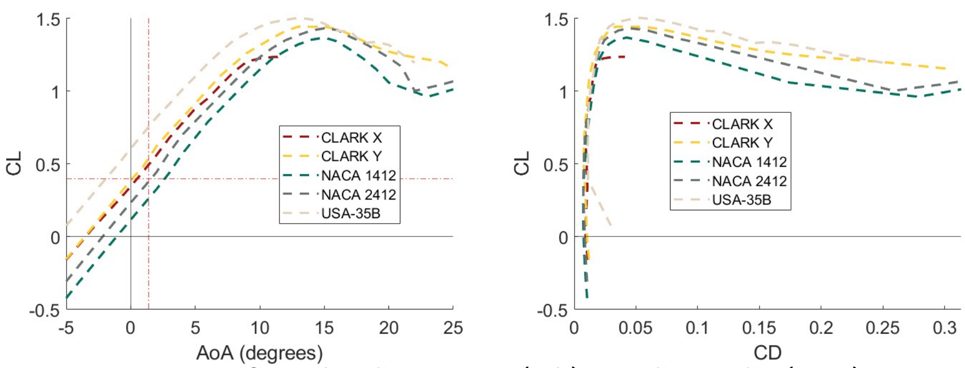


Figure 4.3: Airfoil C_L vs. AoA (left) and C_L vs. C_D (right).

As shown in Fig. 4.3, the USA-35B possessed the highest lift coefficients and widest drag bucket. While the lift generated from this airfoil would improve takeoff (T-O) performance, it would also produce excessive lift at the target cruise velocity of 22.5 m/s, meaning the aircraft would have to fly slower resulting in longer lap times. Similarly, the Clark Y airfoil would also generate more lift than necessary at the target cruise velocity. While the NACA 2412 did not have the highest lift coefficients, it was found to provide sufficient lift at T-O and appropriate lift to meet cruise requirements. The NACA 2412 also had a drag bucket which comparable to the Clark Y. Thus, the NACA 2412 airfoil was chosen because it had the lowest C_{Dmin}, a wide drag bucket, and the desired lift characteristics at the target cruise velocity.

4.4.2 Empennage Sizing

Stabilizer airfoils are generally symmetric as they may either provide lift or downforce, depending on the final trim state of the aircraft. WUDBF selected the NACA 0010 airfoil as it is common for stabilizers and used successfully in past WUDBF projects [1]. Initial sizing of the empennage for stable flight began by investigating the vertical tail and horizontal tail volumes for aircraft of the same class. For a single-engine aircraft, the initial estimate was set at around $\mathbf{V}_{HT} = 0.80$ and $\mathbf{V}_{VT} = 0.04$ [1]. Initial horizontal and vertical tail areas were then solved for using Equation 4.2 and 4.3 [1]:

$$S_{HT} = \frac{V_{HT} \cdot S_{ref} \cdot c_{ref}}{l_T} \tag{4.2}$$

$$S_{VT} = \frac{V_{VT} \cdot S_{ref} \cdot b_{ref}}{l_T} \tag{4.3}$$

Where S_{HT} is the horizontal tail area (cm²), V_{HT} is the horizontal tail volume, S_{VT} is the vertical tail area (cm²), V_{VT} is the vertical tail volume, S_{ref} is the main wing area [cm²], c_{red} is the geometric chord (cm), b_{ref} is the span of the main wing [cm], and I_T is the optimal tail arm length (cm). Based on these calculations, the VT and HT planform areas were set at 353.2 cm² and 767.7 cm² respectively. An initial assumption was made that the AR of the HT would be equal to about half the AR of the main wing and that the AR of the VT would be about 2.5 [1]. Initial estimates for the span of the VT and HT were then made using Equation 4.4 and 4.5 as shown below [1]:

$$b_{HT} = \sqrt{AR_{HT} \cdot S_{HT}} \tag{4.4}$$

$$b_{VT} = \sqrt{AR_{VT} \cdot S_{VT}} \tag{4.5}$$

Where b_{HT} is the horizontal tail span (cm), and b_{VT} is the vertical tail span (cm). Figure 4.4 shows the preliminary configuration of *United Bearlines* compared to the idea aircraft used by Gundmundsson in his calculations.

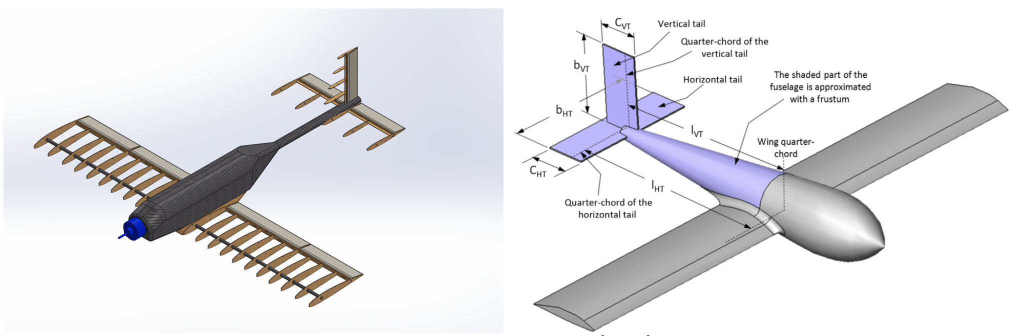


Figure 4.4: Initial configuration of the United Bearlines (left) vs. simplified aircraft layout used by Gudmundsson (right) [1].

4.4.3 High Lift Devices

It was decided that there was no need for high lift devices on the ends of the wing. The cost in weight and drag that high lift devices would add would outweigh the benefit in additional lift that they would give the aircraft.

4.4.4 Control Surface Sizing

Given the traditional design of this aircraft, all control surfaces (rudder, elevator, and ailerons) were designed using rule-of-thumb parameters from Gudmundsson [1]. Control surface sizing was verified using Equation 4.2 (see below) from Gudmundsson [1], previous WUDBF aircraft, and prototype performance.

$$C_{l_{\delta_\alpha}} = \frac{c_{l_{\delta_\alpha}} c_R}{Sb}\left[(b_2^2 - b_1^2) + \frac{4(\lambda - 1)}{3b}(b_2^3 - b_1^3)\right] \tag{4.6}$$

Where $C_{l_{\delta_\alpha}}$ is the aileron authority, $C_{l_{\delta_\alpha}}$ is the change in lift coefficient with aileron deflection, b is the wing span [m], S is the wing area [m^2], c_R is the root chord [m], and λ is the taper ratio. Plain flaps were chosen to simplify the manufacturing process. The hinge point of the flaps was placed at the center of the leading edge arc to minimize large gaps and overhangs when the surfaces are deployed. The flaps were given a chord fraction of 30%, and a span of 40% of the wing. The ailerons had a chord fraction of 30% and a span of 60% of the wing. Both the flaps and the ailerons were designed to deploy between $\pm$ 20°. Using data taken from XFLR5, Fig. 4.5 shows the plotted lift coefficient versus AoA with the flaps deployed at 30° and with no flaps. Chord fractions for the control surfaces on the empennage were 30% for the elevator, and 40% for the rudder. The rudder was designed to deflect a maximum of 30°, while the elevator was designed to deflect a maximum of 30°.

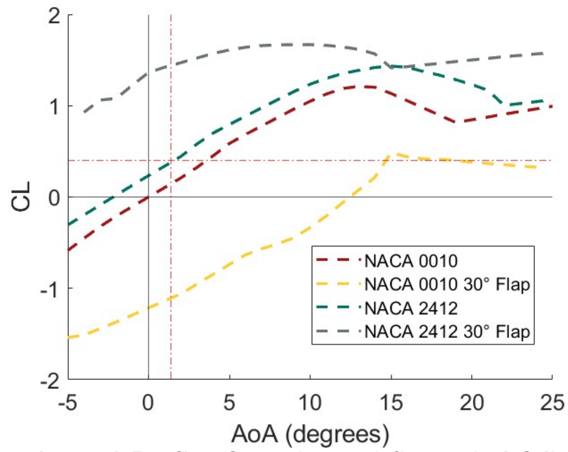


Figure 4.5: C_L of regular and flapped airfoils.

4.4.5 Static Stability Analysis

The static stability of the aircraft is important because it will prevent the aircraft from entering stall and crashing, as well as reorienting the aircraft to fly at the optimized pitch angle of 0°. Good static stability will also help stabilize the aircraft against sudden wind gusts parallel to the flight path.

To verify our aircraft design's static stability, the static margin was calculated using the Equation 4.7.

$$SM = \frac{X_{NP} - X_{CG}}{MAC_{Wing}} \times 100\% \tag{4.7}$$

Where SM is the static margin (%), X_{NP} is the neutral point of the aircraft in the x (longitudinal) direction [cm], X_{CG} is the center of gravity of the aircraft in the x direction [cm], and MAC_{Wing} is the mean aerodynamic chord of the wings [cm]. X_{NP}, X_{CG}, and MAC_{Wing} were automatically calculated by XFLR5. The static margin at cruise for M1 is 11%, and 9% for M2 and M3. This passes well for static stability as a static margin between 5% and 15% is recommended [3]. The overall nature of the static margin and how it changes at extreme angles is shown in Fig. 4.6. It shows that the static margin (proportional to the slope of the lines) is relatively constant at all angles, except for M2 and M3 past 13°. This is acceptable as 13° is an extreme angle that is unlikely to be used, as our takeoff angle is approximately 8°.

4.4.6 Dynamic Stability Analysis

The dynamic stability characteristics of the aircraft were analyzed using XFLR5. The Root Locus plot below in Fig. 4.7 shows the dynamic stability modes for all missions. Based on the weight of the cargo, the stability of M3 is nearly identical to M2 stability. This is ideal as it prevents confusing the pilot between those 2 missions.

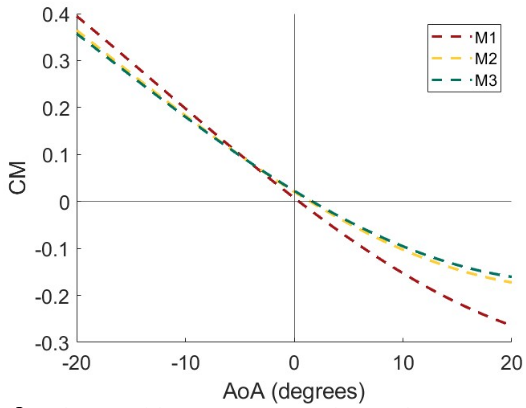
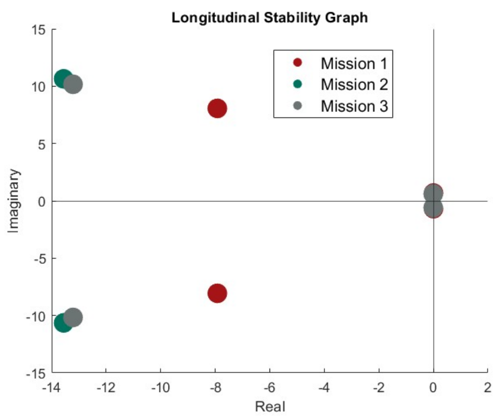


Figure 4.6: Static stability representation using plot of C_M vs. AoA.



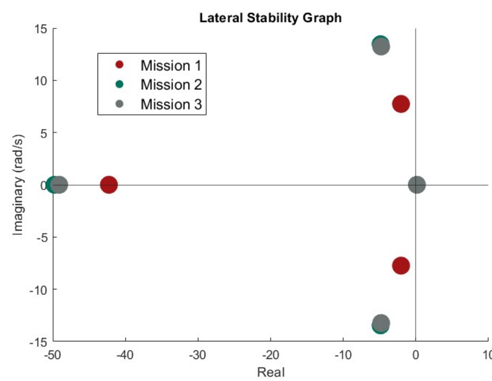


Figure 4.7: Root locus longitudinal plot (left) and lateral plot (right).

Based on the root locus plots, all modes were stable for both loading conditions. The design goal of this aircraft was to satisfy the level 1 flying qualities of a Class 1 aircraft described in MIL-F-8785C [4]. The dynamic stability characteristics from XLFR5 for all missions are shown in Tables 4.3, 4.4, and 4.5.

Table 4.3: Dynamic stability table M1.

Mode	ζ	$\omega_n(rad/s)$	$\zeta\omega_n(rad/s)$	τ (s)
Short Period	0.701 $(0.35 < \zeta < 2.00)$	11.303	8.061	-
Phugoid	0.001 $(\zeta > 0.004)$	0.660	0.660	-
Roll	-	-	-	0.024 $(\tau < 1.4)$
Dutch Roll	0.225 $(\zeta > 0.17)$	7.780 (> 1.0)	7.716 (> 0.35)	-
Spiral	-	-	-	4.944

Table 4.4: Dynamic stability table M2.

Mode	ζ	$\omega_n(rad/s)$	$\zeta\omega_n(rad/s)$	τ (s)
Short Period	0.787 $(0.35 < \zeta < 2.00)$	17.235	10.644	-
Phugoid	0.014 $(\zeta > 0.004)$	0.584	0.584	-
Roll	-	-	-	0.020 $(\tau < 1.4)$
Dutch Roll	0.336 $(\zeta > 0.17)$	14.269 (> 1.0)	13.440 (> 0.35)	-
Spiral	-	-	-	4.971

Table 4.5: Dynamic stability table M3.

Mode	ζ	$\omega_n(rad/s)$	$\zeta\omega_n(rad/s)$	τ (s)
Short Period	0.793 $(0.35 < \zeta < 2.00)$	16.676	10.166	-
Phugoid	0.015 $(\zeta > 0.004)$	0.591	0..591	-
Roll	-	-	-	0.020 $(\tau < 1.4)$
Dutch Roll	0.336 $(\zeta > 0.17)$	14.037 (> 1.0)	13.220 (> 0.35)	-
Spiral	-	-	-	4.751

With the exception of the phugoid damping ratio and the short period natural frequency, all modes satisfied the MIL-F-8785C level 1 requirements [4]. The phugoid damping ratio did satisfy MIL-F-8785C level 2 requirements and pilot evaluation during test flights. The short period frequency fell just outside of level 2, but satisfied pilot evaluation during test flights

4.4.7 Drag Analysis

Ansys returned a force value for drag. The drag coefficient was then calculated using Equation 4.8:

$$C_{dmin} = \frac{2 \cdot F}{\rho \cdot u^2 \cdot A} \tag{4.8}$$

where F is the simulation drag force on that surface, ρ is the air density, u is the cruise velocity, and A is the area of the wings. This yielded a total parasitic drag coefficient of 0.0817. The full parasitic drag breakdown is presented in Fig. 4.8. The component drag build-up method (CDBM) as described by Gudmundsson [1] was not used because it had given inaccurate numbers on previous WUDBF designs. The CDBM method predicted a significantly lower drag coefficient than our flight testing results for previous years.

The wings were the largest contributor to parasitic drag across all mission configurations, which is good, because it means the stabilizers and other components that do not contribute to cargo capacity are not overly significant sources of drag.

The parasitic drag was then added to the lift induced drag, calculated using XFLR5, to give the total drag polar, as shown in Fig. 4.9.

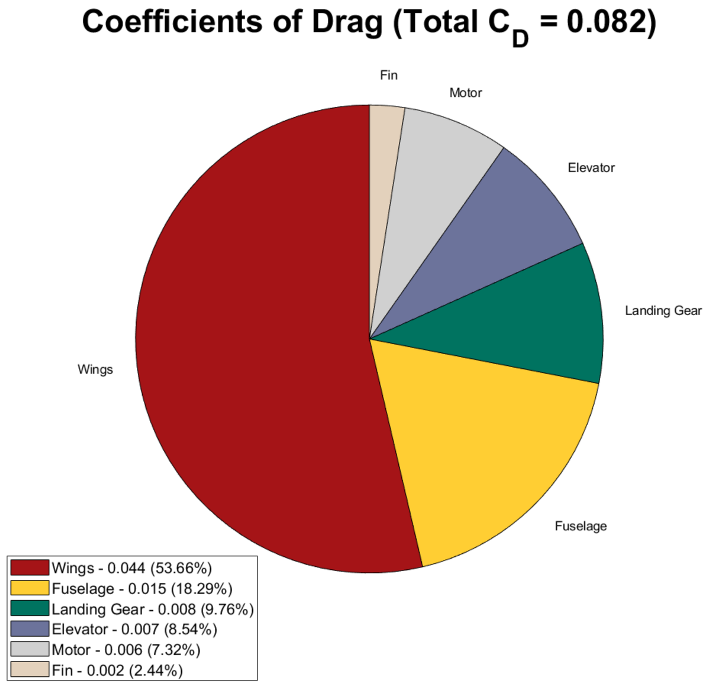


Figure 4.8: Breakdown of parasitic drag by component.

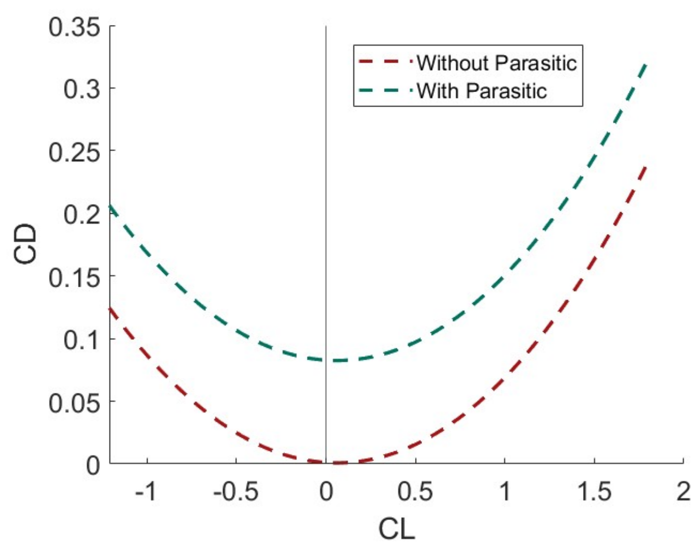


Figure 4.9: Aircraft C_D vs. C_L.

4.5 Propulsion

4.5.1 Battery Selection

The AIAA DBF rules permit the use of either LiPo or NiMH batteries for propulsion. NiMH batteries are generally more robust and safe, but LiPo batteries excel in terms of energy density, discharge rate, efficiency, and dependability. As a result, WUDBF decided to utilize LiPo batteries for their propulsion system.

The propulsion team chose 8s LiPo batteries over 6s for two main reasons: higher power available at takeoff and lower amperage draw. First, the 8s battery has higher voltage than 6s, 29.6V vs. 22.2V respectively, which allows the system to draw less current for the same power output. This is due to the direct relationship between power

(P), voltage (V), and current (I), as given by equation 4.9.

$$P = V * I \tag{4.9}$$

Lowering the amperage draw is favorable as it lead to less heat being generated by the system. This is explained by Joule's first law, which states that the heat generated in a system is proportional to the square of the current, the resistance of the wire, and the time the current flows. The decrease in heat generation leads to increased efficiency as less energy is turning to heat and more is going to power the system.

Second, the higher voltage of an 8s battery allows for higher available thrust and speed when compared to a 6s battery of similar watt-hours. A motor's no-load revolutions per minute (RPM) is given by the equation 4.10

$$RPM = Kv * V \tag{4.10}$$

where Kv is the motor's rotations per voltage supplied (generally given by the manufacturer). So on the same motor and same propeller, an 8s battery could produce higher RPM's than a 6s battery, and thus making more thrust and speed available.

For the battery selection, WUDBF chose to go with the largest capacity battery allowed (see section 3.3.2). The MaxAmps 3250 mAh 8s battery pack was chosen to run the propulsion system because it met the capacity requirement and was readily available.

4.5.2 Motor and Propeller Selection

To find potential motor-propeller combinations, WUDBF utilized the online propulsion design tool, eCalc [5]. With the parameters specified in the following table, WUDBF searched through the eCalc database for motor-propeller combinations that satisfied the stated performance criteria.

Table 4.6: eCalc search parameters.

Parameter	Value	Reasoning
Cruise Velocity [m/s]	27.5	See Section 3.3.1
Endurance [min]	5	See Section 3.3.1
T/W	1.03	See Section 4.2
Battery	8s, 3250 mAh	See Section 4.5.1 & 3.3.2
Max Prop Diameter [in]	14	Estimated max ground clearance

From this search, the propulsion team selected configurations from a set of motor manufacturers trusted by WUDBF that were available, in-stock, at the time of the search and that met or exceeded the requirements in Table 4.6. The propulsion team also took care to allow some margin of error for the flight time by choosing configurations with about 1 additional minute than required. The same was done with the speed as WUDBF noted from previous years that a majority of previous flights were done at around 70% to 80% throttle. This narrowed the possibilities down to the four motor-propeller combinations shown in Table 4.7.

Table 4.7: Motor-propeller combinations.

Outputs	SunnySky X4130-520 V3	Scorpion SII-4025-520	SunnySky X4120-550 V3	SunnySky X3530-520 V3
Propeller	13x6	13x5.5	13x6	13x6
Max Velocity [m/s]	33.61	31.94	33.06	33.06
Mixed Flight Time [min]	6	6.3	5.8	6.2
RPM (static)	13183	13241	13652	13005
T/W	1.93	1.78	1.89	1.87
Static Thrust [kgf]	6.16	5.70	6.06	6.00
Current @ Max [A]	69.94	63.06	71.65	66.53
Drive Weight [kg]	1.244	1.194	1.128	1.141

All four configurations met or exceeded the stated requirements. Therefore, in order to further evaluate each combination, plots of thrust vs. throttle, thrust vs. electrical power, speed vs throttle, and endurance vs speed were generated using data from eCalc:

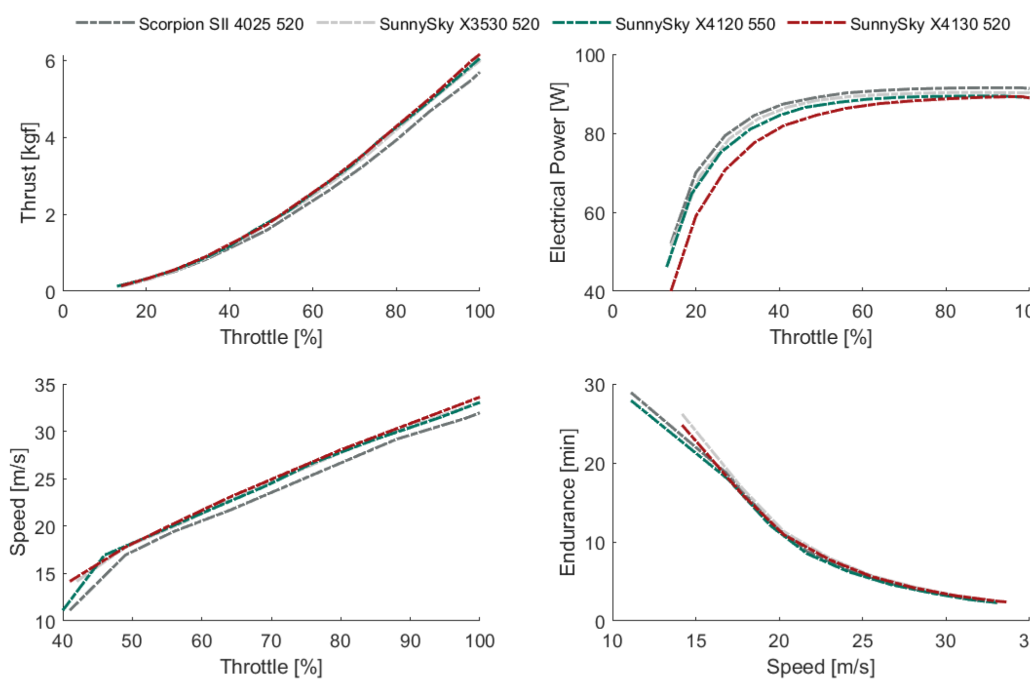


Figure 4.10: Predicted performance data for motor-propeller combinations.

From Fig. 4.10, the SunnySky X4130-520 motor stands out among the other configurations by providing higher thrust and speeds at all throttle levels. Despite its lower efficiency compared to other motor-propeller combinations, the SunnySky X4130-520 compensates for this by reaching the required thrust and speed levels at a reduced throttle levels. This effectively negates its inefficiency, resulting in comparable endurance to the more efficient motors, as seen in the endurance vs. speed graph. Based on these factors, the SunnySky x4130-520 with a 13x6 propeller was ultimately chosen for the aircraft's propulsion system.

4.5.3 ESC and Fuse Selection

Given the predicted max current of 70A for the chosen propulsion system, WUDBF chose to use the Phoenix Edge Lite 100A (8S) as the ESC. The Phoenix Edge Lite can handle 100A continuous current, which provides an adequate safe margin. Additionally, the electronic speed controller (ESC) can handle the increased voltage from the 8S LiPo battery present in the propulsion system.

4.5.4 Refinement for M2

Given that the propulsion system was designed primarily with M3 in mind, WUDBF next considered whether it could be refined to improve M2 performance (since the competition rules allow for the propeller to be swapped out for different missions).

Since M2 has a greater emphasis on speed rather than efficiency, WUDBF considered using a larger propeller with a higher pitch. The larger diameter allows the propeller to move more air, thereby generating greater thrust, allowing for an increase in max velocity. Additionally, the higher pitch moves more air backwards per revolution, contributing to top speeds. The main downside of using larger propellers with a higher pitch is the higher power draw from the motor, leading to lower flight times. Also, higher pitch propellers sacrifice efficiency at lower airspeeds, negatively impacting static thrust and takeoff performance [6]. Because of concerns for the motor power limit, propeller RPM limits, and takeoff distance, WUDBF decided to use the 13x6 propeller for use in all missions.

4.5.5 Required vs. Available Thrust Analysis

The dynamic performance of the propulsion system was evaluated to verify eCalc's predicted performance and to determine the aircraft's speed envelope. eCalc does not output dynamic thrust data so this data was instead used from the APC Propellers website (the chosen propeller manufacturer) [7]. The APC performance data was generated based on vortex theory using actual propeller geometry and the NASA Transonic Airfoil Analysis Computer Program to estimate section lift and drag. Predicted RPM values from eCalc were used to query the APC dataset, with the corresponding dynamic thrust values calculated using an interpolation script written in MATLAB.

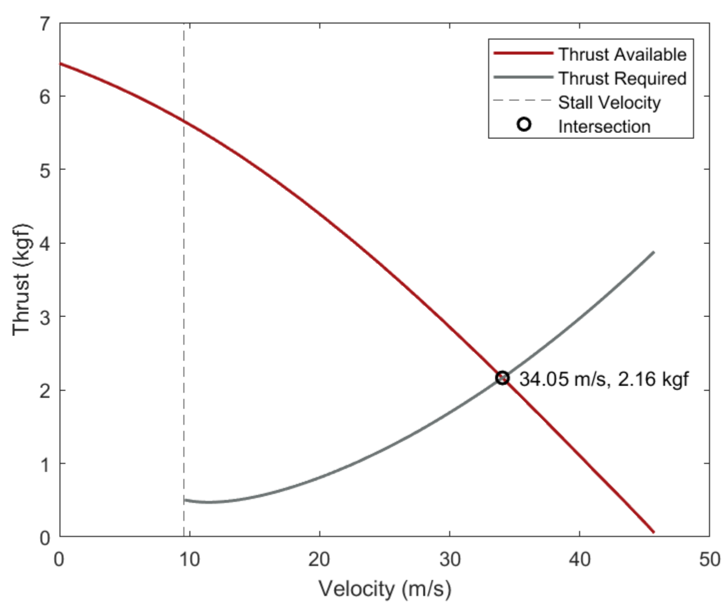


Figure 4.11: Required vs. available thrust curves.

Based on the analysis, the estimated maximum velocity for the aircraft is 34.05 m/s.The velocity calculated above may differ from the value given by eCalc because propCalc (the calculator used to acquire the values from Table 4.7) uses multiple simplifications and assumptions for the drag on the aircraft, which may lead to a difference in the calculated speed vs. the actual speed. The analysis done above more accurately takes into account the drag on the aircraft and thus has a more accurate estimate for the cruise speed of the aircraft.

4.6 Predicted Aircraft Performance

In order to integrate the various aspects of the preliminary design and predict overall aircraft performance, WUDBF utilized Km_sim, an in-house kinematic mission simulator written in MATLAB [8].

4.6.1 Brief Explanation of Km_sim

Km_sim works by simulating an aircraft's performance as it flies through a mission profile. Each mission profile consists of a series of functions based on simple kinematics and Newton's second law describing the motion of the aircraft during a particular phase of flight (e.g., takeoff, climb, cruise, etc.). Specific mission profiles were written for M1, M2, and M3. Aerodynamic and propulsion data from Sections 4.4 and 4.5, respectively, were used to generate aerodynamic and propulsion performance look-up tables. These tables allow the simulation to determine the lift, drag, and thrust of the aircraft at any particular time point. During a simulation, Km_sim steps through each mission phase, numerically integrating the relevant equations of motion using ode45 and the look-up tables, while keeping track of changes in the state variables. At the end of the simulation, plots are generated displaying how the state variables change over time as shown in Fig. 4.12 and Fig. 4.13. For a complete explanation of Km_sim and the source code see [8].

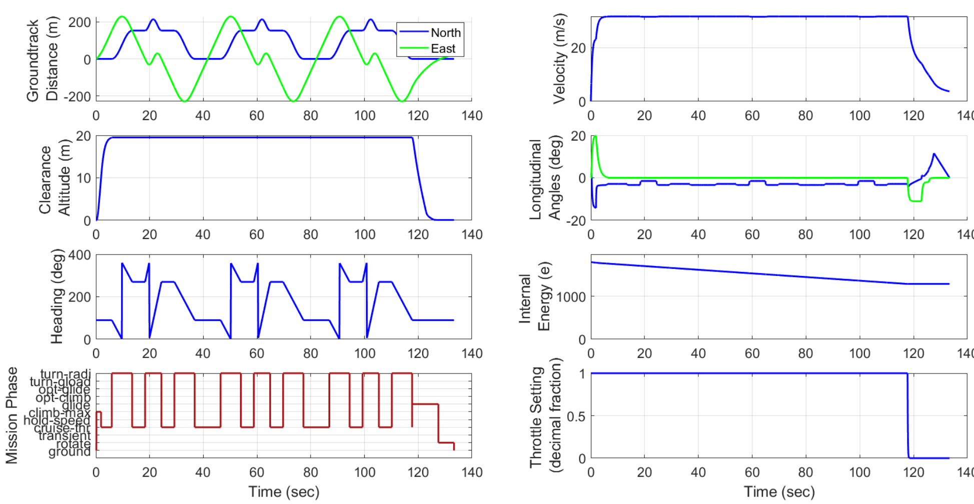


Figure 4.12: Plots of different state variable outputs versus time during a simulation.

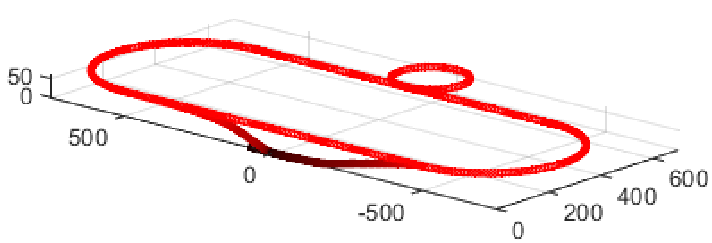


Figure 4.13: Trajectory plot (TOP) illustrating the path taken by the aircraft in 3D coordinates during a simulation.

4.6.2 Km_sim Results

Separate simulations for M1, M2, and M3 were carried out in Km_sim to estimate performance in competition. Normalized scores were calculated based on the maximum competition scores predicted as part of the analysis in Section 3.3. The results of this analysis are shown in Table 4.8.

Table 4.8: Predicted aircraft performance.

	Mission 1	Mission 2	Mission 3
Inputs			
All-Up Weight	2.021 kg	3.651 kg	3.171 kg
Payload Weight	0.250 kg (counterweight)	1.880 kg	1.400 kg
Wing Loading	5.614 kg/m^2	10.142 kg/m^2	8.808 kg/m^2
$C_{L,max}$	1.76	1.76	1.76
Parasitic Drag	0.0817	0.0817	0.0817
Cruise Throttle	1	1	1
Turn Load Factor	2.5	2.5	2.5
Outputs			
Takeoff Distance	4.09 ft	11.87 ft	9.27 ft
$C_{L,cruise}$	0.4	0.4	0.4
$C_{D,cruise}$	0.01	0.01	0.01
Cruise Velocity	34.08 m/s	34.04 m/s	34.06 m/s
Number of Laps	3	3	7
Mission Time at End of Final Lap	1.95 min	1.96 min	4.45 min
Scoring			
Normalized Score	1	1.585	2.214

4.7 Uncertainties

The design tools and analysis methods used throughout the preliminary design process all possessed a certain level of uncertainty. The lift and induced drag of the aircraft was computed in XFLR using the vortex lattice method (VLM). Using VLM, lifting surface thickness and air viscosity were neglected. Additionally, the fuselage was excluded from the analysis. Based on previous WUDBF designs, however, it was determined that the decreased lift due to neglecting viscosity is approximately equal to the lift gained by excluding the fuselage. It has also been verified that the contribution of drag associated with wing thickness was small compared to total drag calculations. Therefore, the lift and induced drag of the actual aircraft was expected to be approximately equal to the XFLR predicted lift using the VLM. The parasitic drag estimate using Gudmundsson's method were also subject to a certain level of uncertainty, but that uncertainty was mostly accounted for CFD analysis. eCalc and the APC propeller performance database were the two primary tools used for propulsion design. These third-party tools both carry a certain level of uncertainty owing to the fact that it is not entirely clear how they calculate their outputs. However, this uncertainty is mitigated by the fact that this method of propulsion design has been used successfully by both WUDBF and other DBF teams in the past with an acceptable level of accuracy. Finally, Km_sim, while a very useful tool, makes a number of assumptions in its performance calculations. For instance, the simulation assumes that the pilot flies the aircraft on a perfect trajectory. The effects of wind are also neglected in the simulation. Since Km_sim relied on data from the aerodynamic and propulsion designs, any uncertainties from those processes were carried over into the simulation. Ultimately, the nature of the iterative process meant that the uncertainty associated with any particular design tool was mitigated by ensuring that every aspect of the design agreed with the rest.

5 Detail Design

5.1 Dimensional Parameters

Following the completion of the preliminary design, the structure and external geometry of the aircraft were finalized during the detail design stage. Table 5.1 summarizes the key dimensions of the final aircraft design.

Table 5.1: Key dimensions of final aircraft design.

Vertical Stabilizer		Horizontal Stabilizer	
Span (mm)	237.8	Span (mm)	520.0
Mean Chord (mm)	156.8	Mean Chord (mm)	210.2
Area (m^2)	0.0373	Area (m^2)	0.1092
Aspect Ratio (AR)	1.52	AR (Aspect Ratio)	2.48
Wing		**Overall Aircraft**	
Span (mm)	1340	Length (mm)	1262
Mean Chord (mm)	270.1	Width (mm)	1340
Area (m^2)	0.3619	Empty Weight (kg)	2.234
Aspect Ratio (AR)	4.96	Max Weight (kg)	3.527

5.2 Structural Characteristics and Design Methodology

WUDBF's primary structural design goals were to maximize strength and stability, maximize accessibility to the electronics and payload compartments, minimize weight, and simplify the manufacturing process. The overall structure is designed to direct loads from each subsystem into the major load-bearing components of the aircraft. The fuselage consists of a fiberglass monocoque body, with a carbon fiber boom running from nose to tail. Propulsive loads from motor thrust and torque that can cause bending, torsion, and vibrations are directed through a wooden motor mount and distributed into the fuselage. Ground loads from the landing gear are directed through the fiberglass-reinforced foam wing box in the fuselage and boom. These load paths are shown in Fig. 5.1 The wing consists of wooden ribs connected by a stringer and a carbon fiber spar, which provides bending and torsional stiffness. Aerodynamic loads from the wing are transferred to the wing box, where they are then distributed across the fuselage structure and the central CF rod. The empennage is directly connected to the central CF rod via a CF spar and locking wood structure, distributing loads into the central rod and the fuselage. Figure 5.1 describes the load path.

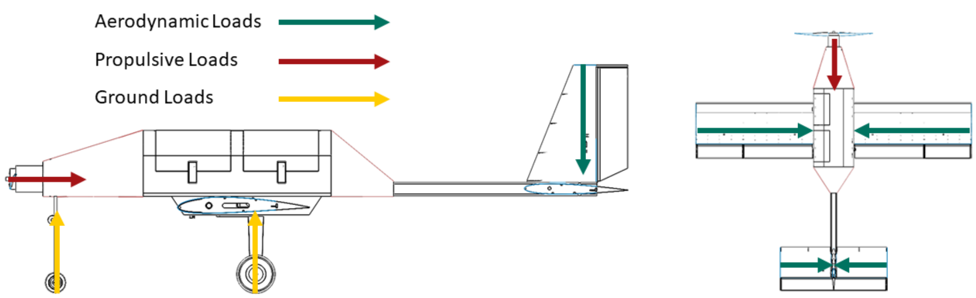


Figure 5.1: Load path diagram

5.3 Subsystem Design

5.3.1 Fuselage Design

The fuselage is a fiberglass monocoque shell, with birch plywood firewalls glued inside. It is primarily designed to encase the electronics package, control electronics, and battery. Also, it directly connects the motor, landing gear, wing, and tail. This design can be seen in Fig. 5.2. Balsa wood plates are used to connect to the motor mounting plate and create shelves for the electronics and battery to sit on. Two openings measuring 6 inches wide are cut into the side of the aircraft. Hinged hatches cover these openings and are secured with plastic latches. The fiberglass fuselage connects a carbon fiber tail boom to the empennage. A foam box is attached underneath the carbon fiber fuselage, allowing for the connection and rotation of the wing.

Figure 5.2: Fuselage.

5.3.2 Wing Design

The wing structure is made up of a total of 22 ribs with one CF spar running through the entire wing. The center portion of the CF rod is attached to the wing box for structural support and to help resist twisting of the wing. The innermost rib of the half-wings have tabs to attach to the sidewalls with nylon bolts. These bolts allow for easy attachment and removal of the alternative wing sections. The wing is able to be rotated 360 degrees under the fuse so it can be parallel with the rest of the fuselage and easily fitted into a box. Servo and control surface attachment points were also included in the wing structure. The control surfaces are constructed from XPS foam with a fiberglass layup. Finally, the leading edges of the ribs are wrapped with balsa wood to increase the stiffness of the wing structure. The wing assembly, showing ribs, CF rod, leading edge, and control surfaces, is shown in Fig. 5.3.

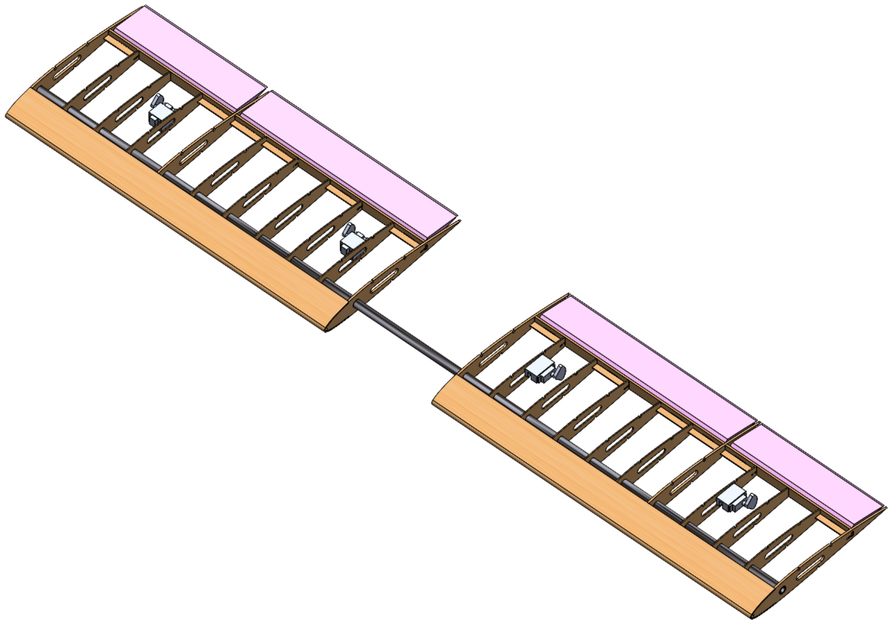


Figure 5.3: Wing assembly.

5.3.3 Empennage Design

The empennage was designed similarly to the main wing design. The horizontal stabilizer is supported by a single CF rod through the entire span. The vertical stabilizer does not have a CF rod. Instead, a plywood tab extends through its lower ribs and fits around the horizontal CF rod. Both the horizontal and vertical stabilizers are glued to the carbon fiber tail boom coming from the fuselage, which increases its stiffness. Mounting plates for control servos and the rudder and elevator hinges are also integrated into the rib structure. This structure can be seen in Fig. 5.4.

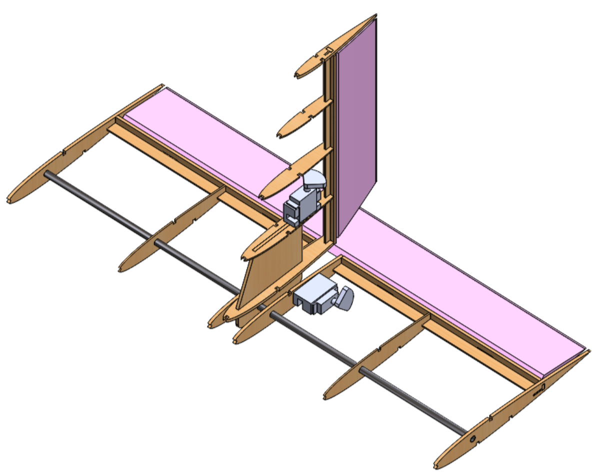


Figure 5.4: Tail assembly showing interlocking between the horizontal and vertical stabilizers.

5.3.4 Wing Rotation

To fit within the 2.5 ft wide parking space, the aircraft must be put into its parking configuration. On *United Bearlines*, shortening the wingspan to 2.5 ft is accomplished by rotating the wing about the vertical axis at a 162° angle relative to the direction of flight. The key component in this subsystem is the 3D-printed wing bearing, shown in Fig. 5.5. The shaft of the bearing is hollow to allow the wing wiring to pass to the fuselage, and the top of the shaft is glued to the boom with epoxy. The bottom of the shaft extends below the wing box, and is flared to prevent the wing box from moving vertically. Additionally, the bearing shaft is glued to the rear landing gear, which allows the landing gear to remain in place while the wing rotates, ensuring the aircraft remains stable during the ground mission. The bearing sleeve fits around the shaft, and its external surface is glued to the wing box. In the flight configuration, loads are borne by four nylon bolts that connect the wing box to the fuselage, so the bearing does not need to be a structural element. These bolts are removed to put the aircraft in the parking configuration.

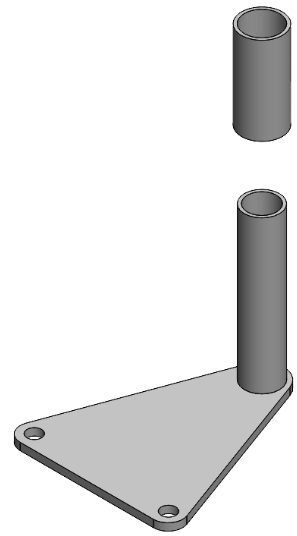


Figure 5.5: Wing rotation bearing showing the shaft (bottom) and the sleeve (top).

5.3.5 Landing Gear

CF landing gear was purchased online because WUDBF did not have the equipment to manufacture it. The JTEC RADIOWAVE JTC-E10, Bow Tricycle configuration landing gear was chosen and placed in the rear of the fuselage. The landing gear is bolted to the bottom of the wing box, which is in turn bolted to the fuselage. This structure is shown in Fig. 5.6. The nosewheel is attached to a shock-absorbing metal rod, which fits into a collar screwed to the motor firewall. A servo in the fuselage allows the pilot to steer the aircraft using the nosewheel.

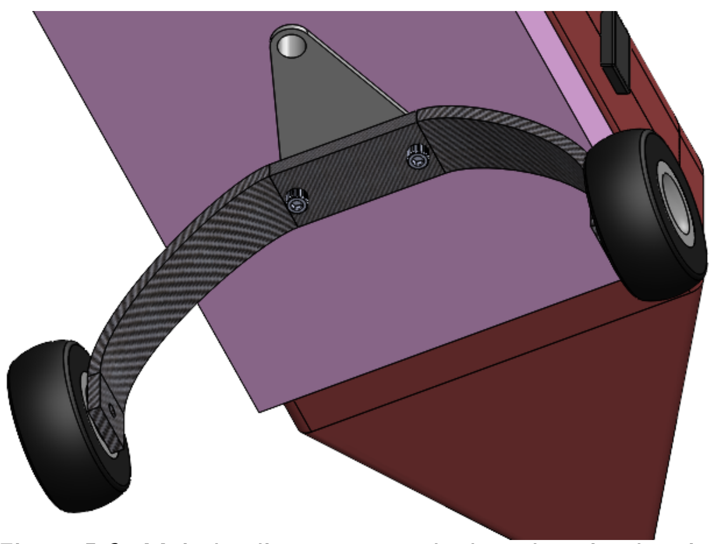
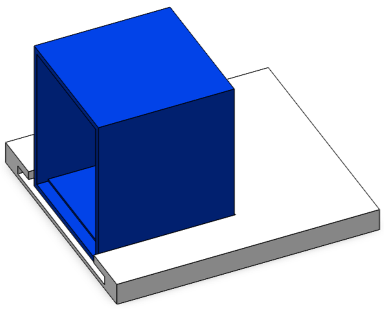


Figure 5.6: Main landing gear attached to the wing bearing.

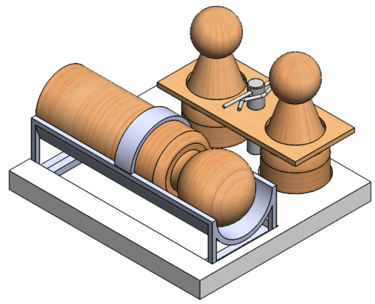
5.3.6 M2 Payloads

There are two different inserts used in M2: one for the Medical Supply Cabinet and one for the Patient, EMTs, and Gurney. The cabinet is secured to its insert using a slotted channel in the insert and a compliant snap fit.

The EMTs are secured to the second insert in the same manner as the passengers, and the gurney slides and snaps into channels in the insert. A tight friction fit secures the Patient fit into the gurney, and a base at the bottom of the gurney allows the patient to not slide out. Both inserts can be seen in Fig. 5.7.



(a) Medical Supply Cabinet and insert.



(b) Patient, EMTs, Gurney, and insert.

Figure 5.7: M2 Inserts.

5.3.7 M3 Payloads

There are two identical passenger inserts, one to be inserted in each hatch. The passengers are arranged on a 3D-printed baseplate in a 4x3 configuration. A locking plate is secured to the insert using cotter pins, and stops the passengers from moving during flight. The insert is attached with velcro to the floor, allowing rapid insertion and removal of the payload. This assembly can be seen in Fig. 5.8.

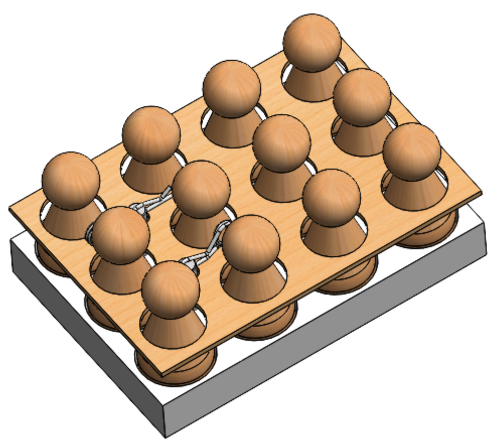


Figure 5.8: Passenger insert.

5.3.8 Crew Insert

The crew insert is constructed almost identically to the EMT and passenger inserts. The insert is permanently affixed to the aft part of the forward compartment, behind the battery, and in front of the firewall. This setup is shown in Fig. 5.9.

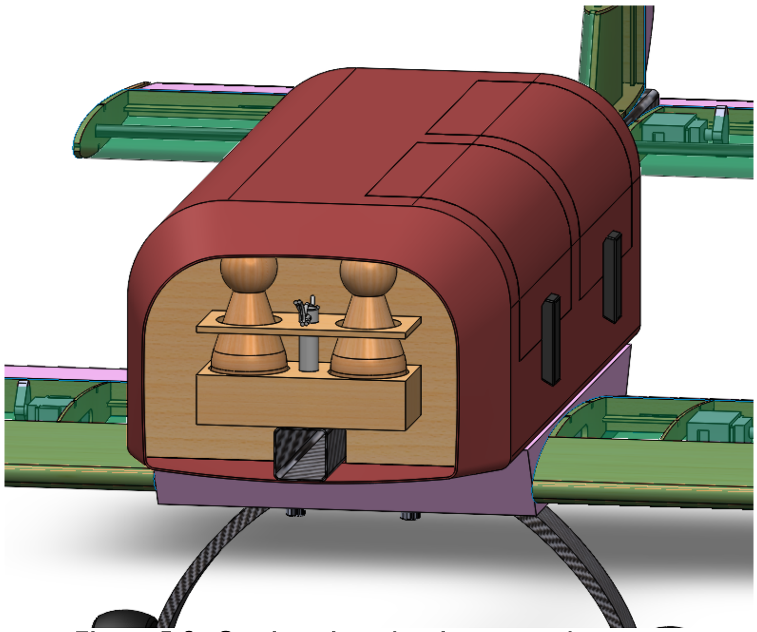


Figure 5.9: Section view showing crew placement.

5.4 Weight and Mass Balance

Each mission has a different mass distribution. To estimate the CG of the vehicle, component masses and locations were determined from the CAD model and tabulated. The leading edge of the imaginary central wing rib was taken to be the origin, and a mutually orthogonal coordinate system was established with the positive X-axis pointing out the nose of the aircraft, the positive Z-axis extending below the aircraft, and the positive Y-axis directed out of the right-wing when viewing the aircraft from behind. The masses and absolute coordinates of the aircraft's major components are listed in Table 5.2. The weights and CG measurements for M2 and M3 were calculated including the payloads for each mission, respectively. Note that components such as bolts, glue, covering, and wire mass are not included in the table. Additionally, the CG for M1 was determined to be in a suitable location, so the counterweight has been removed.

Table 5.2: Weight and balance for each mission.

Component	Weight [kg]	CG_x [mm]	CG_y [mm]	CG_z [mm]
Wing	0.202	-162.46	0.00	-3.56
Fuselage	0.613	-215.09	0.31	-67.39
VT	0.009	-893.31	-0.06	-121.40
HT	0.047	-857.60	-0.02	-38.12
Rear Landing Gear	0.162	-237.83	0.00	76.15
Front Landing Gear	0.005	167.45	-6.83	53.40
6x Control Servos	0.156	-455.33	-5.94	-33.24
Control Electronics	0.100	-353.87	31.69	-34.86
Propulsion Battery	0.415	37.30	0.00	-76.06
Motor	0.475	213.90	0.00	-57.86
M1				
Overall	**2.242**	**-109.44**	**1.07**	**-40.15**
M2				
Gurney and EMTs	0.335	-293.79	31.25	-76.89
Medical Supply Cabinet	1.630	-113.12	30.51	-73.64
Overall	**4.207**	**-125.52**	**14.88**	**-56.05**
M3				
Passengers	1.292	-203.20	0.07	-80.10
Overall	**3.535**	**-143.72**	**0.70**	**-54.76**

5.5 Structural Performance

In order to verify structural performance of the aircraft, WUDBF utilized Solidworks FEA to simulate the major structural components of the aircraft under their maximum expected load conditions.

5.5.1 Motor Forces

The motor forces are a significant source of stress on the aircraft, especially at static conditions such as takeoff. Based on Table 4.7, the static thrust and torque were determined. This was then applied to the motor mount and fuselage. The results are shown in Fig. 5.10.

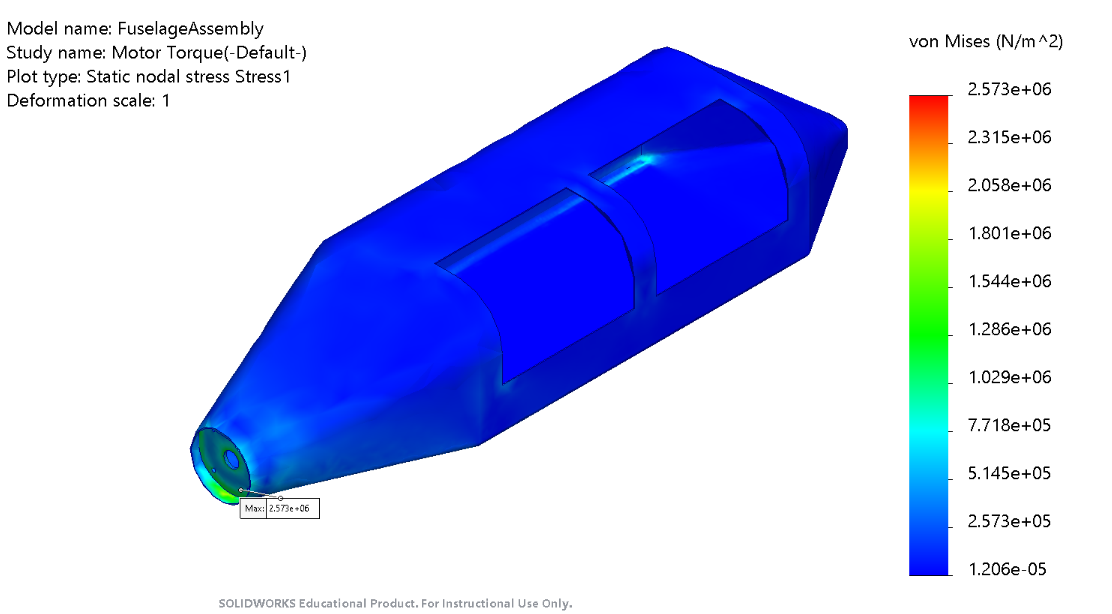


Figure 5.10: von Mises stress plot on the fuselage and motor mount.

The maximum stress was found to be 2.57 MPa, located at on the bottom of the fuselage where it meets the motor mount. This is within the yield limits of both fiberglass and plywood so that the aircraft will remain intact during takeoff [9].

5.5.2 Wing Load on Fuselage

Another significant load on the fuselage comes from the lift and weight forces acting on it during a turn. Using the load factor derived from 4.8 and the heaviest weight from Table 5.2, the maximum load was calculated and applied to the lower surface of the fuselage, where the wing box would be attached. The results are shown in Fig. 5.11.

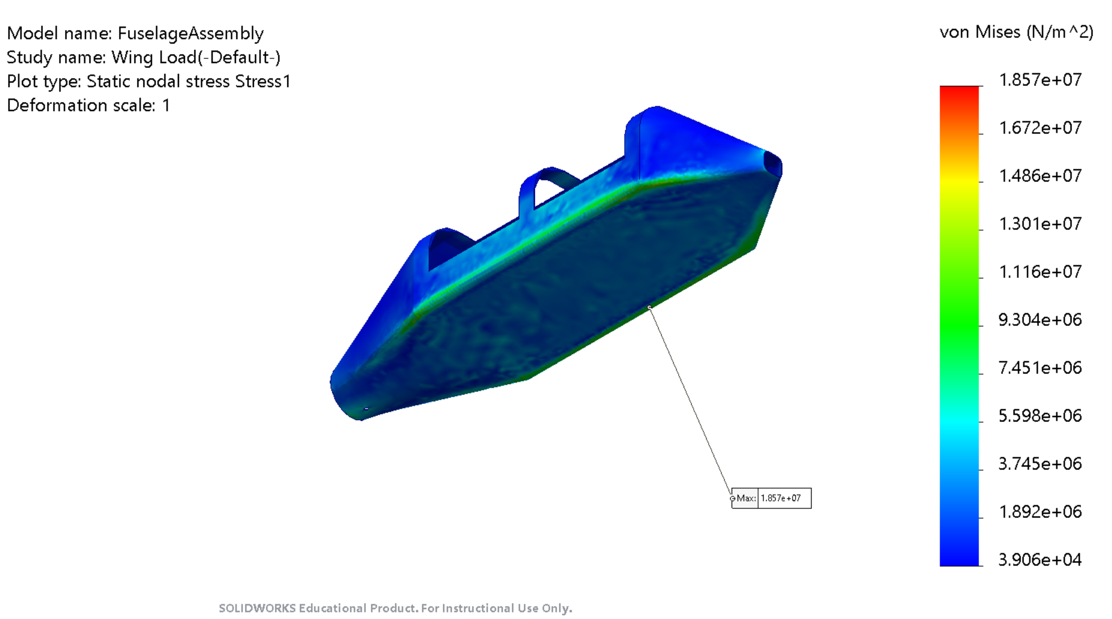


Figure 5.11: von Mises stress plot on the fuselage.

The maximum stress was found to be 18.6 MPa, located at the edges of the bottom fuselage surface. Both the motor and the wing load stress are well below the ultimate stress of 521 MPa for E-Glass fiber, validating the design [9].

5.5.3 Wing Spar FEA

FEA was performed in Solidworks on the CF spar to ensure the wing could withstand a wingtip loading test (approximate load factor of 2.5). The finite lift distribution was approximated by XFLR5 to be near elliptical. In order to input the distribution into Solidworks, curve fitting was performed in MATLAB to generate a continuous distribution as shown in the figure below. Note that the lift distribution is plotted as a fraction of total lift as a function of unit span.

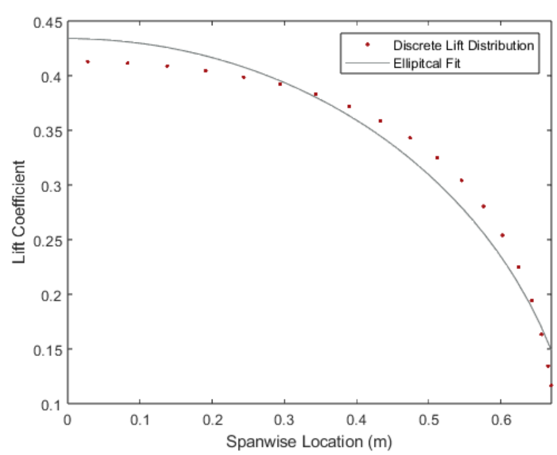


Figure 5.12: Finite and continuous spanwise distribution.

Using the fitted function and a 2.5 load factor, the stress and displacement of the wing was generated. The displacement field is shown in Fig. 5.13.

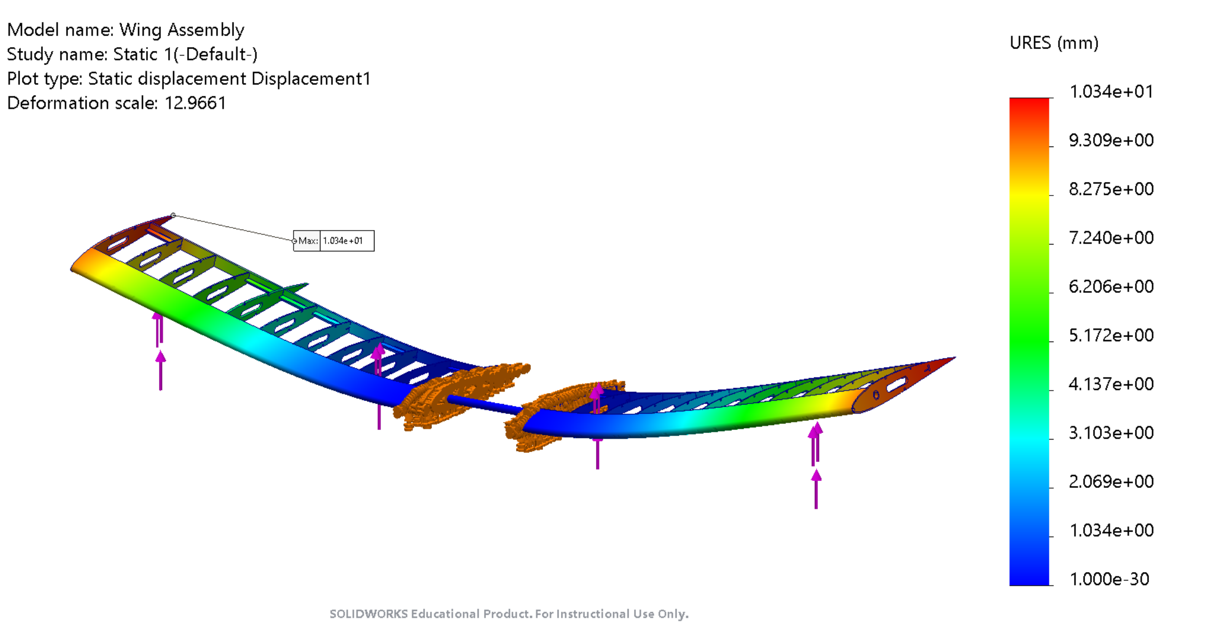


Figure 5.13: Displacement field of wing assembly under load.

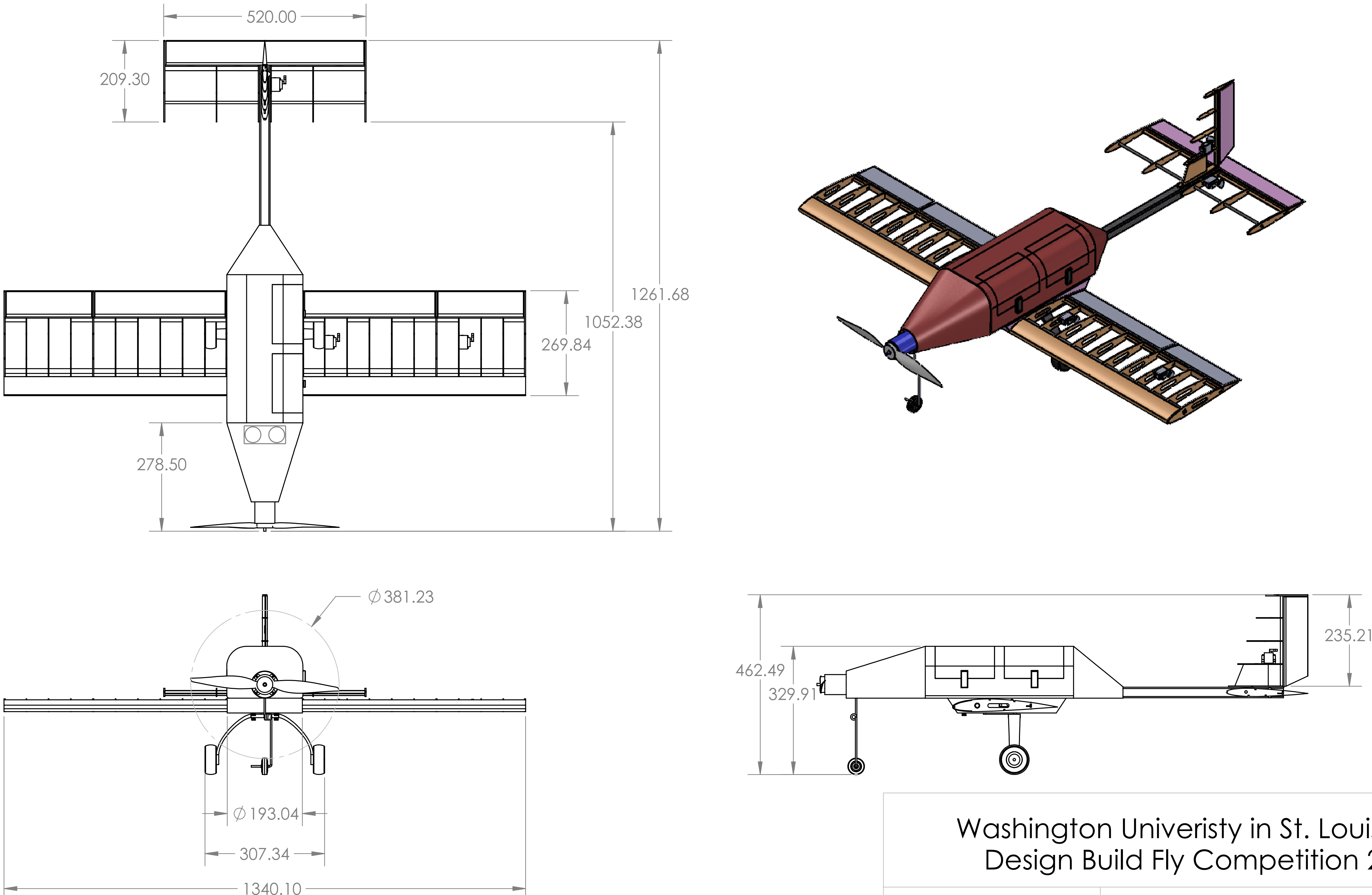
The maximum displacement is 10.3mm at the wingtips. The maximum von Mises stress was found to be 83 MPa, well below the 400 MPa yield stress of intermediate modulus carbon fiber [10].

5.6 Predicted Final Aircraft Performance

Using the updated weights in Table 5.2, Km_sim was utilized to simulate the expected flight and mission performance of the final aircraft. The extracted performance estimates are shown in Table 5.3. To obtain normalized scores, the best scores for M2 and M3 were estimated based on Section 3.3. Note the M2 wing loading is slightly higher than the original target set by the constraint sizing. This change was deemed allowable since the T/W provided by the motor is also higher than the original target.

Table 5.3: Predicted final aircraft performance.

	Mission 1	Mission 2	Mission 3
Inputs			
All-Up Weight	2.242 kg	4.207 kg	3.535 kg
Payload Weight	-	1.965 kg	1.292 kg
Wing Loading	6.228 kg/m^2	11.686 kg/m^2	9.819 kg/m^2
$C_{L,max}$	1.76	1.76	1.76
Parasitic Drag	0.0817	0.0817	0.0817
Cruise Throttle	1	1	1
Turn Load Factor	2.5	2.5	2.5
Outputs			
Takeoff Distance	4.96 ft	15.14 ft	11.22 ft
$C_{L,cruise}$	0.4	0.4	0.4
$C_{D,cruise}$	0.01	0.01	0.01
Cruise Velocity	34.08 m/s	34.01 m/s	34.04 m/s
Number of Laps	3	3	7
Mission Time at End of Final Lap	1.95 min	1.97 min	4.47 min
Scoring			
Normalized Score	1	1.611	2.196



All dimensions are in millimeters

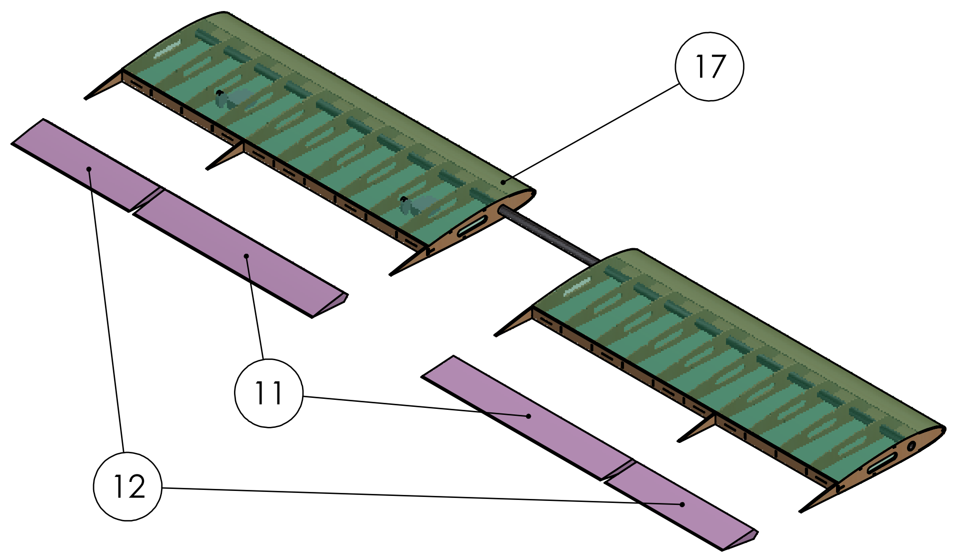
Washington Univeristy in St. Louis AIAA
Design Build Fly Competition 2024

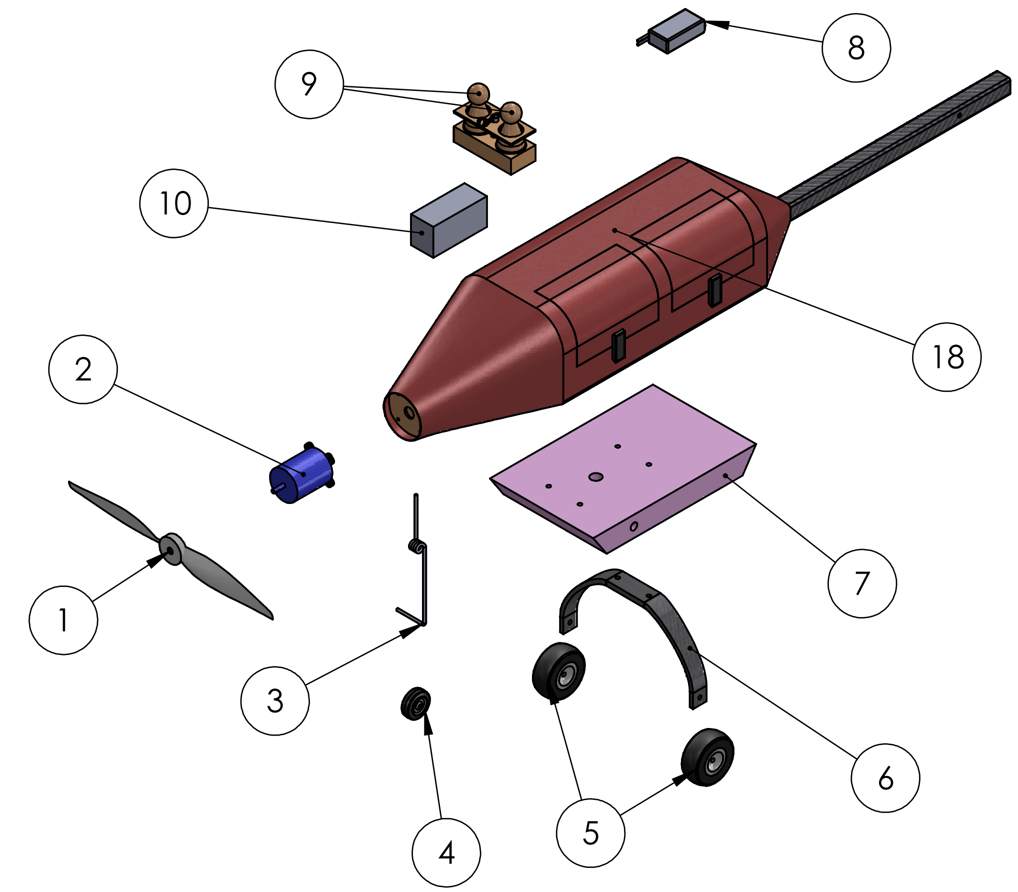
Name:
United Bearlines

Aircraft Detail 3-View

SCALE: 1:10

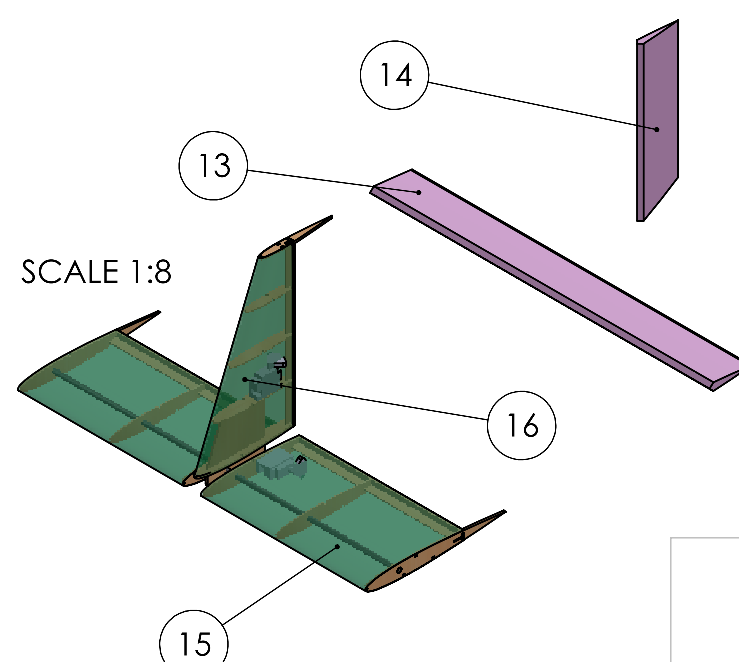
SHEET 1 OF 4





ITEM NO.	PART NUMBER	QTY.	Material
1	Propeller	1	Plastic
2	Motor	1	Aluminum
3	Front Landing Gear Spring	1	Steel
4	Front Landing Gear Wheel	1	Foam
5	Rear Landing Gear Wheels	1	Nylon
6	Rear Landing Gear	1	Carbon Fiber
7	Wing Mount Box	1	XPS Foam
8	GEN Recciver Pack	1	-
9	Crew	2	Birch Wood
10	Battery	1	Lithium Polymer
11	Flaps	2	XPS Foam
12	Ailerons	2	XPS Foam
13	Elevator	1	XPS Foam
14	Rudder	1	XPS Foam
15	Horizontal Stabilizer	1	Balsa Wood
16	Vertical Stabilizer	1	Balsa Wood
17	Wing	1	Balsa Wood
18	Fuselage	1	Fiberglass

SCALE 1:8



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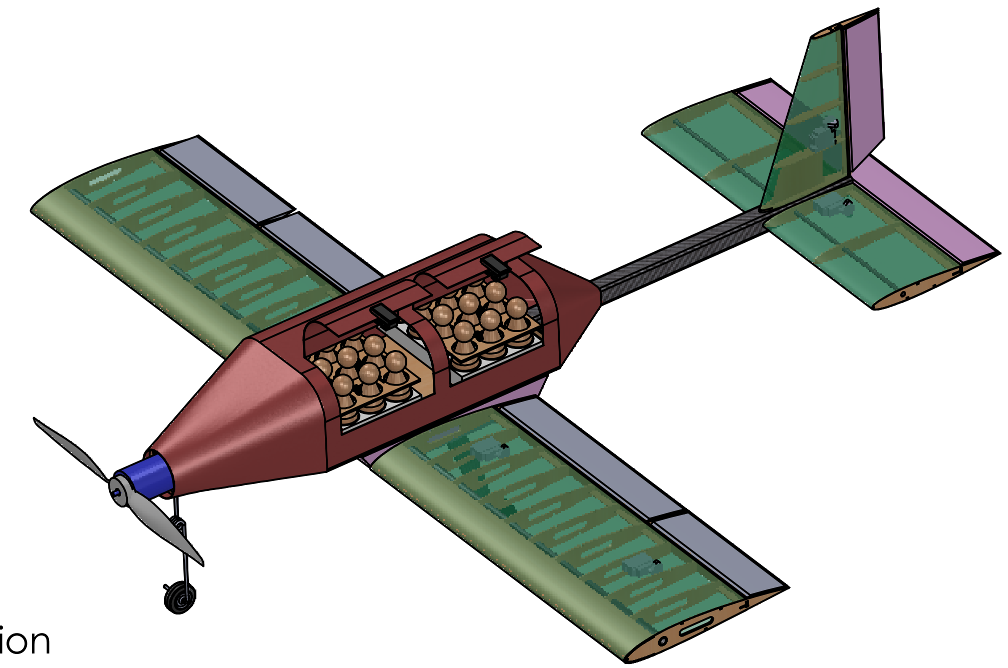
Name:
United Bearlines

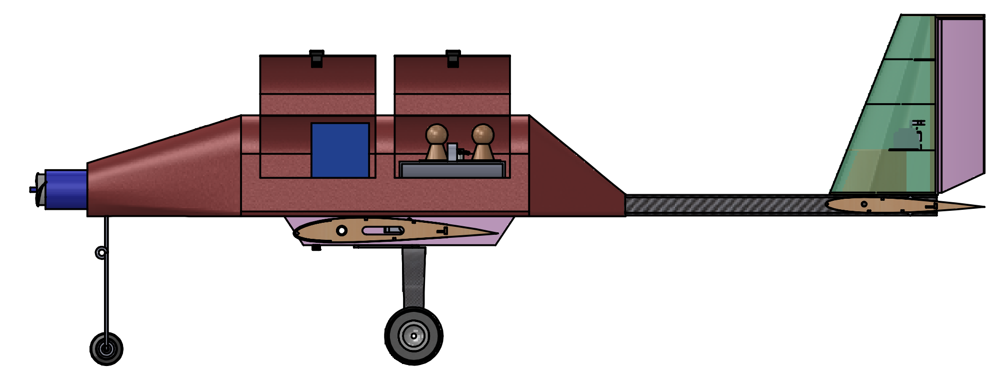
Aircraft Exploded Layout

SCALE: 1:10

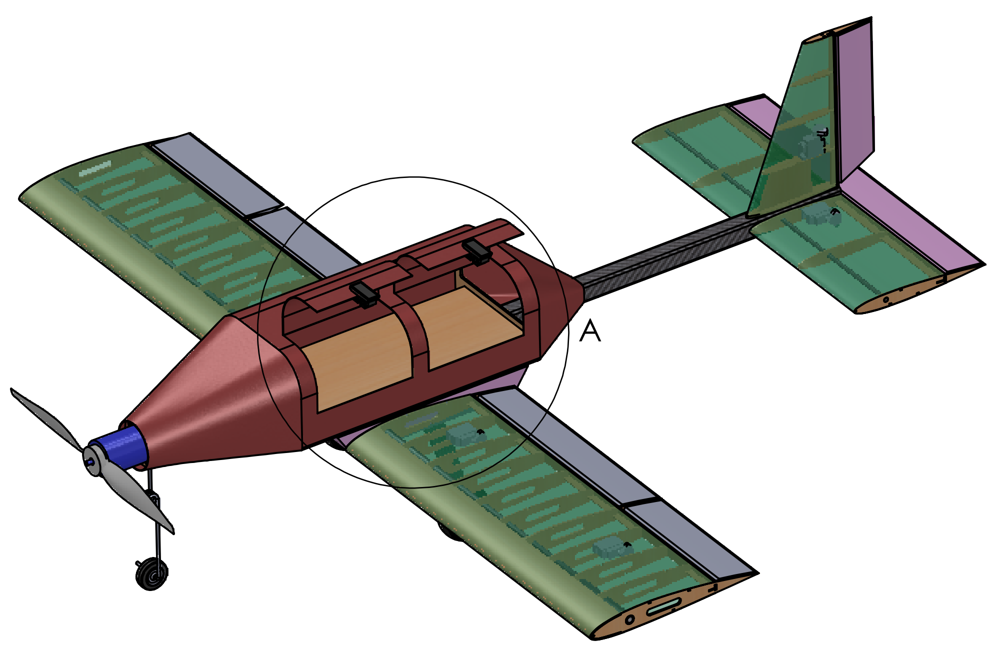
SHEET 2 OF 4



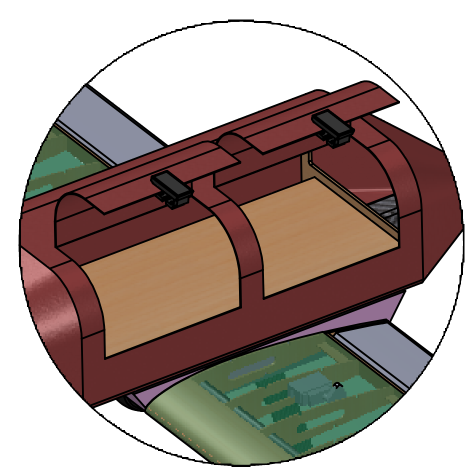
Passenger Mission



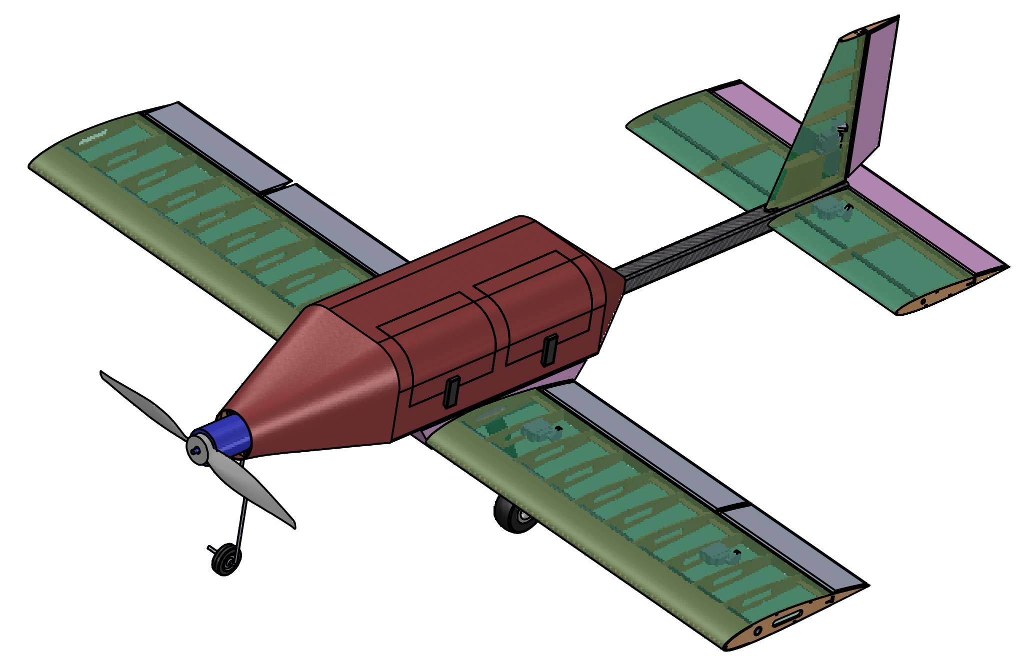
Medical Mission



Hinge to close plane doors

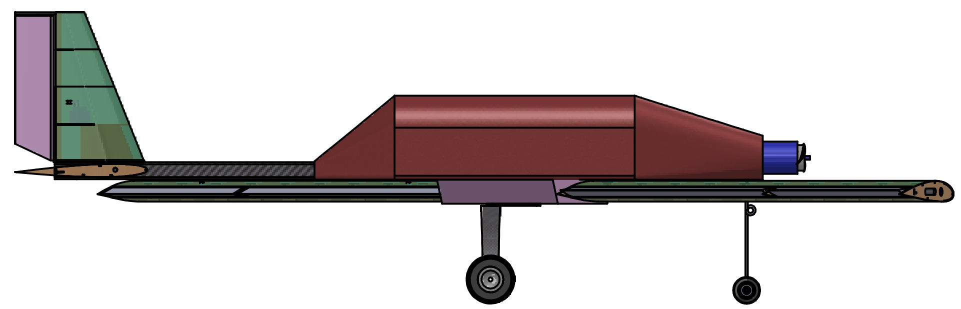


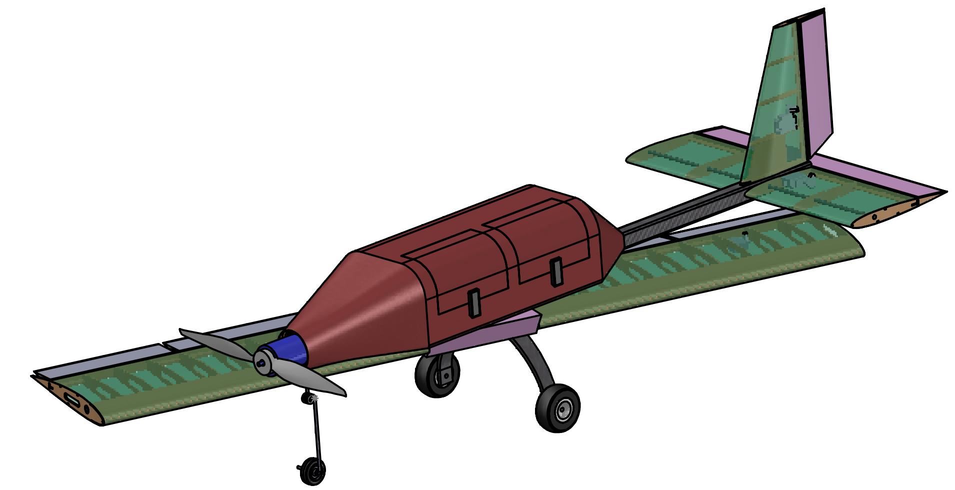
DETAIL A
SCALE 1 : 7



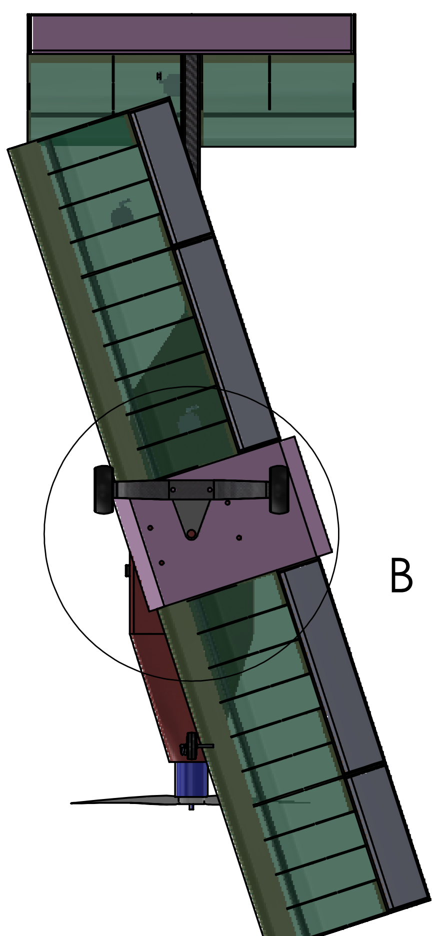
Washington Univeristy in St. Louis AIAA
Design Build Fly Competition 2024

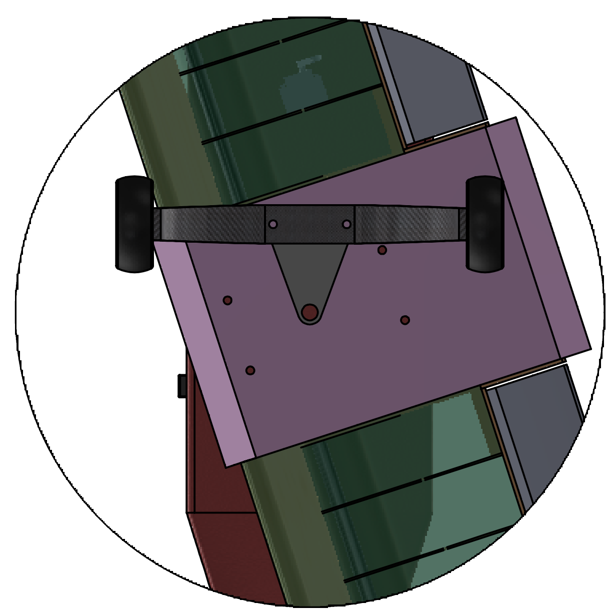
Name: United Bearlines	Mission Accomodations
SCALE: 1:10	SHEET 3 OF 4



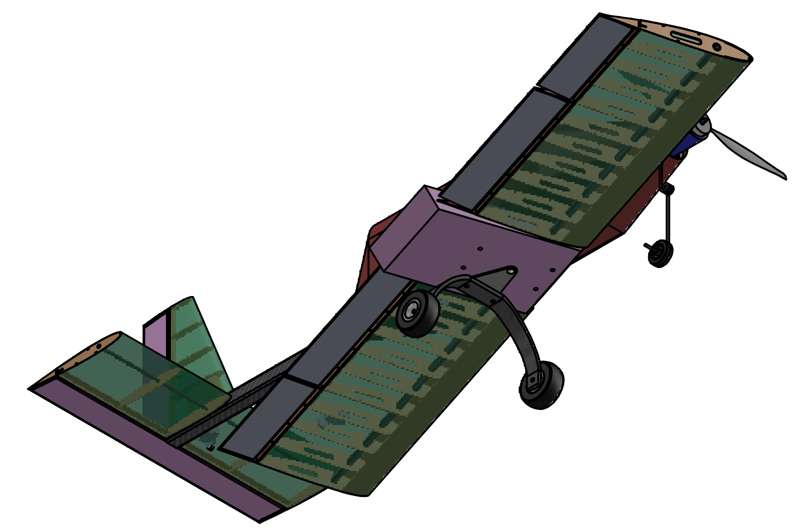


Wing Rotation Mechanism





DETAIL B
SCALE 1 : 6



Washington Univeristy in St. Louis AIAA
Design Build Fly Competition 2024

Name: United Bearlines	Aircraft Wing Rotation
SCALE: 1:12	SHEET 4 OF 4

6 Manufacturing Plan

6.1 Selection of Manufacturing Methods

Several different manufacturing processes were identified to select the best method to construct the aircraft and its numerous subsystems.

6.1.1 Selection Criteria

Key evaluation criteria included ease of construction, effectiveness, feasibility, and weight. Materials that had been used in past aircraft were generally preferred, given manufacturing methods for those materials had already been refined. The cost was also considered due to the club's limited budget.

6.1.2 Solid Foam

Solid foam components can be designed and fabricated quickly since there are no internal bracing structures. Using foam along with carbon fiber and fiberglass sufficiently strengthens the foam. Additionally, the use of a hot-wire cutter to accurately shape the material makes it a desirable material to form smaller, precise parts of the aircraft such as the ailerons, rudder, and elevator. Other advantages of using foam to fabricate parts of the aircraft include its lightweight.

6.1.3 Additive Manufacturing

3D printing has advantages during construction due to the ease of fabricating structures as single components and the streamlined workflow from CAD to printer. However, 3D printers are slower than other methods for constructing large shapes. 3D printed parts are also heavier and weaker than those made from other materials, especially in shear along the seams between layers. While WUDBF ultimately did not make use of 3D printed parts for the main aircraft structure, more intricate components that don't bear significant loads such as servo mounts were 3D printed due to the flexibility and ease of 3D printing. Additionally, 3D printing was used to create molds for composite layups.

6.1.4 Composite Layup

Due to their high strength and low weight, fiberglass composite layup methods were used to construct the fuselage. These methods require the use of molds or plugs, which were 3D printed or made from foam. The composite layups consisted of multiple layers of fiberglass. Epoxy was used to glue the layers together. This fuses the materials which hardens and strengthens the layup. Additionally, prefabricated CF rods and fiberglass layups were used to reinforce the major load-bearing components of the aircraft.

6.1.5 Wood Construction

Wood construction is the manufacturing technique WUDBF is most experienced with. WUDBF has a laser cutter that can be used to accurately and rapidly cut out wooden parts. A number of different woods are used, primarily balsa, basswood, and birch ply. Each type of wood offers different material properties. This creates flexibility when choosing where weight can be saved or where extra strength is needed. Assembly of wooden components can be done using cyanoacrylate (CA) glue and epoxy. CA glue dries quickly and forms a strong bond, making it the primary adhesive in wooden construction. Two-part epoxy offers a stronger alternative for components that bear more structural loads. To maintain the aerodynamic shape of the aircraft, the frame can be covered in Ultracote, a heat-shrinking plastic film that can be tacked onto the aircraft and then heated, pulling it tight across the frame. Ultracote is relatively thin and susceptible to tearing but can be quickly and easily repaired.

6.1.6 Process Selected For Major Components

Fiberglass layups were selected for fuselage construction due to strength properties. Wood construction, with Ultracote coating, was chosen as the primary manufacturing method for most other major components due to its structural qualities and WUDBF's previous experience. Wooden components including wing ribs and tail formers are cut using a laser cutter. These pieces are then assembled using CA and epoxy. CF parts and fiberglass reinforcement are added in key areas for structural support. In particular, the central rod, wing spars, and landing gear are CF parts purchased online. The foam box connecting the landing gear and wings to the fuselage is reinforced with fiberglass.

6.2 Detailed Manufacturing Processes

The manufacturing processes used in the construction of the aircraft are detailed below. Note that these descriptions are primarily based on the construction of the prototype aircraft and some adjustments may occur for the final aircraft, which is currently being built.

6.2.1 Wing and Tail Structure

The wing is composed of ribs, strings, and a main CF spar. Each half-wing is constructed separately. One central CF spar slides through the first half-wing, the foam base, and then the second half-wing. The ribs are aligned using slotted basswood jigs, with each rib fitting into a designated slot. Once the ribs are aligned, the CF spar is inserted and secured using epoxy/CA. A quarter-inch square basswood stringer is then sanded and attached to the leading edge to achieve the correct curve. Next, the mounting plates for the control surfaces are installed, the jigs cut off and sanded down, and the wing flipped over. Basswood stringers are added. The leading edge is then formed by wrapping sheets of balsa soaked in warm water around the leading edge stringer. 3D-printed servo mounts are then glued to the ribs for later electronics installation. Control surfaces are constructed from foam and reinforced with fiberglass. Hinges are then inserted and epoxied into the control surfaces and onto the hinge plates. Finally, the wing and control surfaces are covered with Ultracote.

The tail is constructed using basswood stringers, balsa wood, and thin CF rods. Like in the wing, basswood stringers serve as the leading edges. These stringers are sanded to the shape of the desired airfoils. Each component of the tail is constructed separately in an identical manner to the wings, using similarly designed jigs. The elevator and rudder are also constructed separately and then attached to the HT and VT respectively, using hinges and CA glue. To connect the tail to the fuselage, holes oriented perpendicularly to each other are drilled into the central CF rod, which extends out of the fuselage and acts as the tail boom. A custom hole drilling jig is used to ensure that the holes are perpendicular to each other and aligned properly with the wings and the rest of the aircraft. Next, a thinner CF rod connected to the tail is slid through the square CF rod that serves as the tail boom. Then, the VT and HT are epoxied to the box. After the installation of servos and servo mounts, the tail is covered in Ultracote.

6.2.2 Fuselage Structure

The fuselage is constructed primarily from fiberglass composite layups. The fuselage was split into pieces that were then used to create stencils for mold construction. The hot-wire foam cutter was then used to cut foam into the mold pieces. These pieces were assembled with Elmer's glue. The fiberglass core layers were compressed by vacuum packing. Additional layers were epoxied on after an initial curing period. Inside the fuselage, basswood ribs were used to reinforce the walls for extra strength. Two basswood landing gear plates are then installed under the main fuselage. These plates are reinforced with fiberglass for extra strength. The CF landing gear is sandwiched by these plates and secured using nylon bolts. The nose mount contains the motor and is laser cut from birch ply. It is slid into the fuselage and secured with epoxy. A foam base is epoxied to the bottom of the main fuselage. The wings are on a single plane and are connected with a CF rod. The CF rod goes through the foam base.

2 oz plain weave fabric was chosen for its balance of flexibility and strength. Based on empirical testing (see section 8.1.2), a layup of three plys was chosen. Stacking sequence was decided based on the results of section 5.5. The motor and wing load forces act perpendicularly to each other, so orienting the fibers in a 0°/90° orientation would best resist those forces. However, the motor torque generates shear stress, which would be best resisted by 45° plys. The further a ply is away from the center line, the more that ply resists bending moments. Since the wing loading force is the greatest magnitude, the layup consists of two 0°/90° plys sandwiching a 45°/-45° ply.

6.2.3 Landing Gear

The landing gear is made of CF and prefabricated. Next, the landing gear is secured using nylon bolts and basswood plates reinforced with fiberglass to withstand impact. A tricycle setup was chosen, with the front wheel allowing a small amount of freedom for steering.

6.2.4 Wing Folding

The wing rotation bearing is 3D-printed from resin for its superior strength and smoother surface compared to fused filament fabrication printers. The bearing sleeve is affixed to the wing box with epoxy, and the wing box is reinforced with fiberglass. The bearing shaft is glued to the landing gear. Four nylon bolts inserted from the bottom of the wing box hold the assembly together while in the flight configuration.

6.2.5 Inserts

The insert bases are 3D-printed, while the locking plates are constructed from basswood. Steel cotter pins secure the assembly, which is attached to the fuselage floor with velcro.

6.2.6 Payloads

The Medicine Supply Cabinet is constructed as boxes filled with dense material. The walls of the box are made of basswood. The box is secured with CA glue and the lid is removable so that the weight is changeable. Zinc-plated, ultra-smooth BBs are used as weights in the box. This allows the CG to remain adjustable. The gurney is 3D-printed.

6.3 Manufacturing Milestones

A manufacturing schedule was implemented to ensure that components were constructed on time. The construction of the prototype and final aircraft were divided into important subtasks which were then used to create the schedule shown in Fig. 6.1 below.

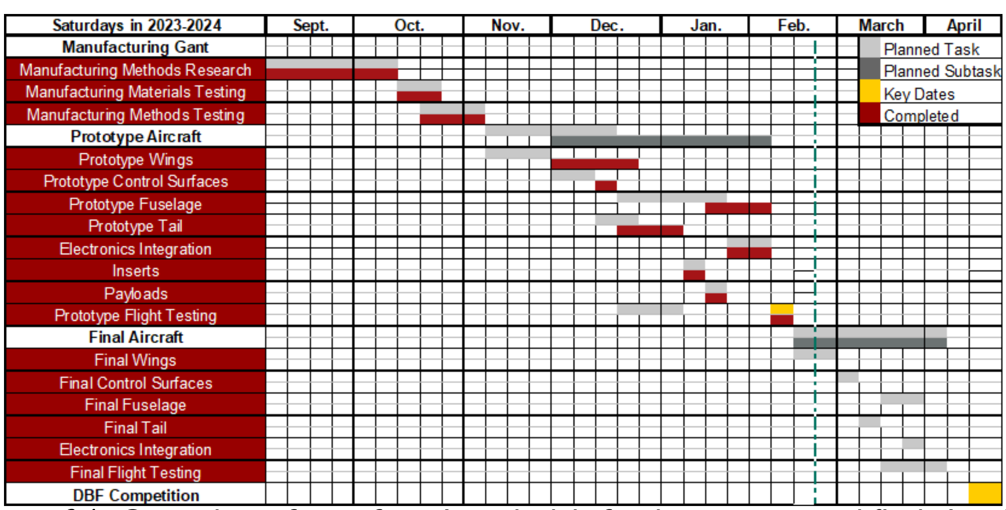


Figure 6.1: Gantt chart of manufacturing schedule for the prototype and final aircraft.

7 Testing Plan

Ground and flight testing of the aircraft were performed to validate the design and drive iterations toward the final aircraft design.

7.1 Overall Test Objectives

Primary test objectives are listed in the table below by category. These objectives were achieved through several ground and flight tests described in Sections 7.2 and 7.3 respectively.

Table 7.1: Breakdown of test objectives.

Category	Objective
Propulsion	Collect data from static thrust tests to confirm propulsion performance on the ground. Validate the propulsion system in the air during flight tests to confirm 20 ft takeoff capability, cruise velocity, and endurance.
Manufacturing	Investigate manufacturing processes, in particular the benefits of composite fuselage construction.
Structures	Conduct wingtip loading tests at maximum AUW to verify structural integrity. Perform torsion and flutter tests to ensure that deflections are not excessive. Confirm the integrity of the landing gear via ground tests.
Takeoff	Confirm aircraft can takeoff within 20 ft.
Ground Mission	Ensure rapid loading and unloading of payloads can be achieved.
Performance	Record flight data for analysis to confirm performance requirements are met. Obtain feedback from the pilot regarding handling qualities.

7.2 Ground Testing

7.2.1 Propulsion Test

Static thrust testing was performed on the chosen propulsion system using an RCbenchmark Series 1585 Dynamometer mounted to a thrust stand. Using RCbenchmark software, thrust, RPM, current draw, and electrical power measurements were collected at incremental throttle levels. The data were then used to validate predicted performance and confirm the propulsion system met the necessary requirements.

7.2.2 Fiberglass Test

This year was the first that WUDBF used a monocoque composite fuselage. With the mission requirements emphasizing a lightweight aircraft, WUDBF needed to find the minimum layup thickness that provided the required strength. This was accomplished by laminating samples of one- through six-ply 2 oz fiberglass. These layups were then weighed and subjected to a bending test and Euler-Bernoulli beam theory was used to calculate their moduli. These values were then compared to the area density of the layup.

7.2.3 Ground Mission Test

Before the prototype was constructed, a mock-up fuselage was created out of foam board, and a door was cut into the side in accordance with the competition rules. WUDBF then simulated the ground mission by loading and unloading passengers into a prototype insert and installing it into the aircraft. These tests both informed the scoring analysis and provided valuable practice for the ground mission crew member.

7.2.4 Wingtip Test

The prototype aircraft was subjected to a wingtip loading test with the maximum payload weight to verify its structural integrity. This test simulated a 2.5g turn and the wingtip test required at competition. Results were also validated against the FEA results from section 5.5.3.

7.2.5 Flight Testing

A test flight was conducted to verify that the aircraft met design goals. A Pixhawk flight computer was used to capture telemetry during each flight test (see figure below). This data was then analyzed to extract measures of aircraft performance including speed, endurance, and lap times. Feedback from the pilot was also used to evaluate the stability and flight characteristics of the aircraft. Conclusions from flight testing were used to inform changes to the design going from the prototype to the final aircraft. For the full details of the flight test, see the test schedule in Table 7.4.

7.2.6 Ground Inspection and Pre-Flight Checklists

Table 7.2: Ground inspection checklist.

Component	Tasks
Motor	☐ Motor secured ☐ Propeller free of damage
Fuselage	☐ No cracks or tears ☐ Electronics are connected and secured
Wing	☐ No cracks or tears ☐ Bolted to fuselage ☐ Wingtip test
Payload	☐ Payload is positioned correctly and secure
Control Surfaces	☐ No cracks or tears ☐ Servos connected to receiver ☐ Control surfaces can deflect fully
Empennage	☐ No cracks or tears ☐ Control surfaces move without interference
Landing Gear	☐ Front wheel secured and servo connected ☐ Rear wheels secured

Table 7.3: Pre- and post-flight checklists.

Flight Checklists
Pre-Flight
☐ Payload secured ☐ Propulsion battery level check ☐ Receiver battery level check ☐ CG position confirmed ☐ Wingtip test ☐ Receiver turned on ☐ Control surfaces test ☐ Landing gear steering test ☐ Range check ☐ Motor run-up ☐ Hatches and payloads are secure
Post-Flight
☐ Throttle down ☐ Arming plug removed ☐ Propulsion battery disconnected ☐ Receiver battery disconnected

7.3 Testing Schedule

Table 7.4: Testing schedule.

Date	Type	Category	Objective
Prototype			
10/14/2023	Ground	Performance	Simulated loading and unloading of passengers.
1/20/2024	Ground	Manufacturing	Fiberglass strength testing.
1/27/2024	Ground	Propulsion	Static thrust test to confirm propulsion performance.
2/14/2024	Ground	Structures	Wingtip test to confirm structure.
2/18/2024	Flight	Performance	Shakedown flight. Test prototype design.
2/18/2024	Flight	Performance	Full M1 simulated flight. Confirm TOFL. (CRASHED).
Competition Aircraft (Planned)			
3/9/2024	Ground	Performance	Ground mission test.
3/30/2024	Flight	Performance	M1 Simulated flight. Evaluate performance.
3/30/2024	Flight	Performance	M2 Simulated flight. Evaluate performance.
3/30/2024	Flight	Performance	M3 Simulated flight. Evaluate performance.
4/6/2024	Flight	Performance	Practice for competition.

8 Performance Results

8.1 Ground Test Results

8.1.1 Propulsion Test Results

Results from the static thrust test confirmed that the propulsion system generally met the desired performance requirements. Test data compared to eCalc predicted performance in the table

Table 8.1: Initial design targets.

Parameter	Prediction	Test Result
RPM	13183	13438
Thrust [kgf]	6.16	6.09
Max Current [A]	69.94	75.50
Electrical Power [W]	1909	2360.9

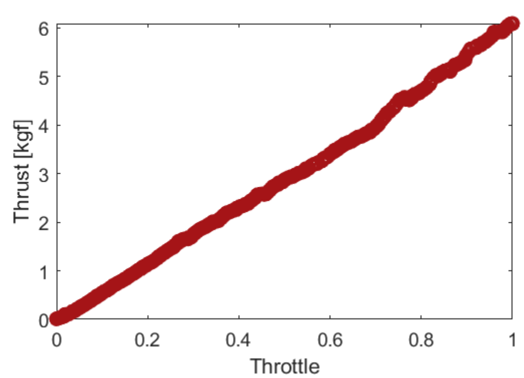


Figure 8.1: Static Thrust Test Results

Based on the static testing, the motor drew slightly more burst current than predicted. This is not an issue, however, since the ESC was sized with significant margin. The efficiency of the actual propulsion system appeared to be slightly lower than predicted, however obstructions and ground effects may have contributed to that decrease in efficiency. Later flight testing confirmed that the propulsion system delivered the desired takeoff capability and cruise performance.

8.1.2 Fiberglass Test

The results of the bending test are shown in Fig. 8.2

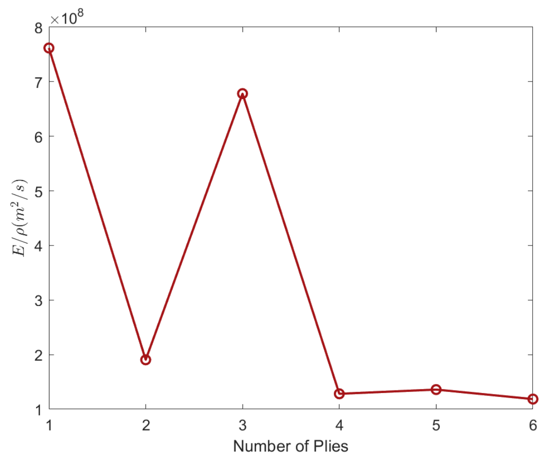


Figure 8.2: Fiberglass test results.

As a result of the testing, it was determined that a single ply had the best strength-to-weight ratio. However, one ply is too flimsy for use on the aircraft. The clear second-best layup was a 3-ply laminate, which offered good stiffness in addition to being relatively lightweight.

8.1.3 Ground Mission Test

Time trials with the mockup fuselage showed the time to load or unload one passenger was around one second. The overhead time required to install and remove each insert was found to be 9 seconds. Using these values in the scoring analysis validated WUDBF's choice to carry around half of the estimated maximum number of passengers to balance GM and M3 scores and maximize the overall score.

8.1.4 Wingtip Test

The aicraft was successfully supported by the wingtips. The deflection was found to be 10mm at the wingtips, which is lower than the predicted value of since the fuselage components add additional structural support.

8.2 Flight Testing

8.2.1 Feb. 18th, 2024 - Multiple Flights (Prototype)

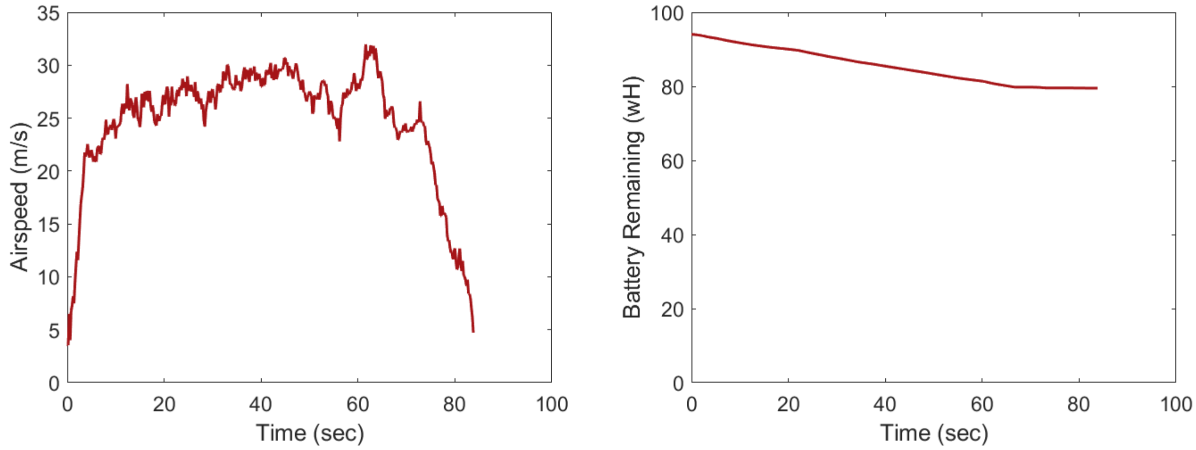
United Bearlines performed two flights during the Feb. 18th test flight of the first prototype. The preliminary design can be seen in Fig. 8.3. Table 8.2 summarizes the outcomes of the Feb 18th test flight.

Figure 8.3: United Bearlines on Flight 2 of Feb. 18 Test Flights

Table 8.2: Test flight outcomes.

Flight Description	Result
Shakedown	The pilot trimmed the aircraft and was able to achieve stable flight characteristics. After the aircraft contacted the ground during landing, its landing gear detached. Despite this issue, the rest of the aircraft stayed intact and the aircraft was able to safely land along the runway. As a result of this shakedown, extra care was put into securing the landing gear so that it would not detach during future flights.
Full Simulated M1	The servo wire connecting the elevator servo disconnected just after takeoff. Once elevator control was lost, the pilot was able to turn the aircraft 90° before crashing. The fuselage remained largely intact, verifying its strength. This failure prompted extra precautions to be put into place so that the elevator servo would stay connected during competition.

Figure 8.4 shows the airspeed and battery remaining throughout the duration of the shakedown. The aircraft reached a maximum airspeed of 31.98 m/s. Only 14.51 Wh of battery were discharged during the 84 second flight, prompting optimism about the aircraft being able to fly for the full 5 minutes in M3.



(a) Airspeed vs. time. (b) Battery remaining vs. time.

Figure 8.4: Airspeed and battery capacity during the shakedown.

Table 8.3 compares the predicted performance with that of the actual prototype.

Table 8.3: Predicted performance vs. actual performance during the shakedown.

Flight Description	Predicted	Actual	Difference
M1 Cruise Velocity	34.08 m/s	31.98 m/s	−6.4%
M3 Battery Discharge Rate	Less than 0.32 wH/s	0.17 wH/s	−61%

References

[1] Gudmundsson, S., 2022, *General Aviation Aircraft Design: Applied Methods and Procedures*, 2nd ed., Elsevier.

[2] "Airfoil database search," accessed 13 September 2023, http://airfoiltools.com/

[3] "cgCalc - Center of Gravity (CG) Calculator," accessed 16 November 2023, https://www.ecalc.ch/cgcalc.php?deeplink=Arcus%20(Beispiel);cm;mono;17.5;17.5;13.5;10.5;6;0;-5;-5;-2;4.5;14;68.5;56;35.5;22.5;3.5;0;0;0;0;0;0;0;0;0;0;0;0;0;0;0;0;0;0;0.80;11.5;5.5;4;0;0;0;4;5;0;0;0;24.5;3.5;0;0;0;92.5;25;10;#:~:text=The%20CG%20value%20is%20always,stability%20is%20called%20Static%20Margin.

[4] Foster, T. M., 2017, "Dynamic Stability and Handling Qualities of Small Unmanned-Aerial-Vehicles UNMANNED-AERIAL-VEHICLES," BYU ScholarsArchive.

[5] "eCalc - Reliable Electric Drive Simulations," accessed 8 November 2023, https://ecalc.ch/index.htm

[6] Raymer, D. P., 2018, *Aircraft Design: A Conceptual Approach*, 6th ed., AIAA.

[7] "APC Performance Data," accessed 20 January 2024, https://www.apcprop.com/technical-information/performance-data/

[8] Hainline, K., "Km_sim," accessed 30 October 2023, https://www.mathworks.com/matlabcentral/fileexchange/94968-km_sim

[9] "E-Glass Fiber - Generic," MatWeb, accessed 15 November 2023, https://www.matweb.com/search/DataSheet.aspx?MatGUID=d9c18047c49147a2a7c0b0bb1743e812

[10] "Properties of Carbon Fiber," Clearwater Composites, accessed 15 November 2023, "https://www.clearwatercomposites.com/resources/properties-of-carbon-fiber/"